A carrier readout for the zeros of ζ, and a machine-checked index-to-ordinate extractor

Samuel Lavery

Draft — September 19, 2026

Abstract

We work in a three-dimensional model and descend to the classical one-dimensional setting only as a readout. The stage is a double-ended Archimedean helix with closed-form arclength; a *carrier* — a numberline scaled by a harmonic constant $\Delta > 0$ — is laid along it, its sites at equal arclength $x_n = \Delta n$; and the L-function rides the carrier as a *fiber*, whose phasors the conductor sorts into channels that cancel *focally*, at a height, leaving no residue. The one-dimensional L-function is this object seen in a chart, reached by dropping the radius (Cayley) and then the angle ($y = \log z$); the height survives, and the classical ordinate is its registration.

In the chart the construction is a one-parameter family of *carrier readouts*: real-valued functions u_Δ on the critical line, the chart of the bank at each height. In the chart that readout is Hardy's Z up to a strictly positive factor, and the isolation results below are what the chart reads: they are stated and proved as properties of the readout, and they coincide with the classical argument, which is the form the chart gives them. §18 sets out what lives only in three dimensions. One item is not available in the chart at all: in the ordinate variable the distance off the critical line is a *radial* coordinate, so a certified contour count over a band $|\beta - \frac{1}{2}| \le h$, matched against certified sign changes, certifies that no zero in a window has left the line by as much as h (Theorem 56), and an elementary certificate on the sliver near $\operatorname{Re} s = 1$ removes the band, so that every zero with ordinate in the window lies on the line (Corollary 59) — a statement about β, which the readout's sign provably cannot reach. We prove, unconditionally and with machine-checked proofs, that the warped fiber on the scaled carrier *is* the one-dimensional function — its ordered chart equals $\Delta^{-s}(1 - 2^{1-s})\zeta(s)$ on the whole channel $\operatorname{Re} s > 0$, on the three-dimensional phasors and at every carrier scale, and determines ζ everywhere by the identity theorem (Theorem 17); that u_Δ is a strictly positive multiple of $\Lambda(\frac{1}{2} + it)$, so its zeros are exactly the on-line zeros of ζ at every carrier scale; that its sign changes by $(-1)^m$ across an event of multiplicity m, with no simplicity hypothesis; that zeros off the line pair at a common ordinate, so the parity of the *full* strip count is the readout's sign change; and that a sign change across an interval on which Ξ' does not vanish forces exactly one event there, simple. Chaining this yields the index: the i-th cell contains the i-th event, uniquely, and a sign-change sub-bracket of that cell encloses a zero of ζ on the critical line. Every hypothesis is an evaluation at finitely many points; no counting routine, zero-density estimate, or bound on $S(t)$ enters.

Separately, and off that path, the *abscissa* of a zero proves recoverable from data computed only on the line: the bank's log-jets are exactly a power sum over the fiber's zeros, a zero $\beta + i\gamma$ becomes a node $\gamma - i(\beta - \frac{1}{2})$, and a matrix pencil returns those nodes as eigenvalues. The reading is calibrated against off-line zeros the fiber carries in closed form.

The development is formalised in Lean 4/Mathlib — 100 theorems in the chart chain, with the geometry, its descent and ten supporting modules besides; zero `sorry`, zero `axiom`. An implementation is reported and validated at 297 indices, and its own hypotheses are

discharged by computation against an explicit Euler–Maclaurin remainder. Two propositions earlier carried as open hypotheses are *disproved* rather than assumed. §19 states precisely what is not claimed.

Contribution. This is a geometric model for the Dirichlet L-functions, and it computes. Its objects are a *scaled carrier* — a numberline reparametrized by a harmonic constant on its own integer lattice, the sites at arclength $x_n = \Delta n$ — and a *warp*, a reparametrization by a function: dynamic, readout-preserving, of unit modulus. The warp's generator is a self-adjoint operator on ℓ^2 with the phasors as eigenbasis; the fiber is its flow, ζ is its trace, and the critical line is the boundary of the Hilbert space in which the fiber lives (§17). The fiber rides the carrier; the lanes are the conductor's $\pm$ channels; the weld gauge is the fiber's own argument, and it absorbs the root number, so the readout is real for every primitive character and not only for the self-dual ones. The model is carrier-generic: reality, the functional equation and the counts are properties of the involution and the readout map, established once for every fiber that rides the carrier, with ζ the case formalised here. From that geometry — addressed by index, with no Gram point and no asymptotic expansion needed to *define* the readout — one receives, given a count satisfying the interface of §13, a certified interval containing the i-th on-line zero, and the chain down to $\zeta(\frac{1}{2} + i\gamma) = 0$ is machine-checked; §19(iv) says exactly how much that certifies. (The Riemann–Siegel formula does appear later, but only as an interchangeable *evaluator* of the readout's sign, §20(O4); nothing in the model or the theorems depends on it.) Geometric readings of ζ have to this point been descriptive; this one is operational.

Status of the obligations. A companion aim of this paper is that it should carry no unproved hypothesis and no unbacked measurement. Every theorem here is unconditional, no theorem cited takes an unproved proposition as an argument, and the implementation's own hypotheses are now discharged by computation: the sign evaluations are certified against an explicit Euler–Maclaurin remainder, the non-vanishing of Ξ' is certified on the intervals returned, and the zero count is certified by contour, all from an explicit region bound on $|\Xi|$ derived here rather than quoted. Two propositions that earlier drafts carried as open hypotheses are *disproved* instead: the μ_6 sector balance law (Theorem 13) and the fixed-cutoff clock bound (Proposition 66). §20 gives the status of each item and what closed it. **Keywords.** Riemann zeta function; zeros on the critical line; certified computation; formalized mathematics; Lean 4; Prony's problem.

2020 Mathematics Subject Classification. Primary 11M06, 11Y35; Secondary 11M26, 68V20.

Code availability. The Lean sources (`RequestProject/`), the implementation and every numerical check named here live in the project repository, and §22 lists what each one runs and the decision rule it was given before it ran. They are *not* embedded in this PDF: a reader sent the PDF alone cannot run any of it, and no claim here should be taken on the strength of a script the reader cannot see.

Contents

1 Introduction **4**

2 The geometry: the double-ended helix **7**

3 The carrier: a numberline, harmonically scaled **8**

4 The fiber: the L-function riding the carrier **9**

5 The descent to the chart **12**

6 The chart readout: lanes and the weld gauge **13**

7 The ordinate variable and the entire readout **16**

8 Off-line pairing and the strip count **18**

9 The interval count **19**

10 A count of one without a count **19**

11 The chain and the capstone **20**

12 The jet enclosure **21**

13 The extractor **23**
13.1 The head law . . . 23
13.2 Algorithm . . . 24
13.3 Validation . . . 25

14 Eigenvalues and the rank drop: reading the abscissa on the line **26**
14.1 The jets are a power sum over the fiber's zeros . . . 26
14.2 The node dictionary . . . 26
14.3 The pencil . . . 27
14.4 What the rank counts . . . 27
14.5 The calibration standard the fiber carries . . . 27
14.6 Lehmer's pair, resolved . . . 28
14.7 The singular values, certified . . . 28
14.8 Propose and verify: the node locations, certified . . . 29
14.9 The rank, certified . . . 29
14.10 The inertia: what the sign of the Hankel spectrum counts . . . 30
14.11 Status . . . 31

15 The model for a general Dirichlet L-function **31**

16 A certified on-line verification, carried out in the ordinate variable **32**

17 The Hamiltonian of the fiber, and the three-dimensional zeta **34**
17.1 The log-weight operator . . . 34
17.2 The warp is the functional calculus of H . . . 35
17.3 The one-dimensional function is the trace . . . 36
17.4 The three-dimensional zeta is the bank, and the chart's zero is an event . . . 36

18 Where the geometry is load-bearing **37**

19 What is not claimed **40**

20 Remaining obligations **42**

21 Formalization **47**

22 Reproducibility **49**

23 What the exercise settled **51**

1 Introduction

What is constructed

The object is a three-dimensional one, and the order of exposition follows it. There are three layers, and they are kept apart throughout because conflating them is what makes the subject look one-dimensional.

The geometry (§2) is the stage: a double-ended Archimedean helix

$$\gamma(k) \;=\; \big(rk\cos 2\pi k,\; rk\sin 2\pi k,\; pk\big)$$

together with its conjugate anti-helix through the origin, both running $0 \to +\infty$. It has a closed-form arclength and no marks. Scaling does not act here.

The carrier (§3) is a numberline laid along the geometry — a ruler — and it is where harmonic scaling acts: the sites sit at equal *arclength* $x_n = \Delta n$ for a harmonic constant $\Delta > 0$. One geometry can host several such rails at once.

The fiber (§4) is what rides the carrier: the L-function's representation, carrying a phasor at each site consumed. Phasors are born at zero magnitude and grow; they spin in the logarithm because the ruler they sit on turns. The conductor sorts them into positive and negative channels — for ζ, the even and odd sites, which is what a conductor of 2 supplies. A *warp* — scaling by a function, unit modulus — moves the fiber through configuration space and sets the height at which its channels cancel. That cancellation is *focal*: the channels point at a common height and close there, exactly, leaving no residue. It is an event at a height, not a root of a formula.

Only then does the one-dimensional L-function appear, and it appears as a *readout* (§5): 3-D $\to$ 2-D by Cayley, dropping the radius; 2-D $\to$ 1-D by $y = \log z$, dropping the angle. The height survives both steps, and the classical ordinate is its registration — the two coincide, and are not the same quantity. The logarithm of the entire model enters once, in the second step. That the geometric event and the analytic one coincide is proved, as an *iff*, for non-principal real characters (Theorem 11); for ζ that particular form degenerates and the licence is instead an exact factorisation of the fiber on the line (Theorem 19), with Remark 12 recording precisely how much of the geometric form is proved for ζ and how much is a named hypothesis.

In the chart (§6 onward) the fiber is a complex number at each height which turns as the height advances; the *weld gauge* is its own argument, read off from the archimedean local factor, and dividing it out leaves a real number — the *carrier readout*. The rest of the paper is statements about where that readout vanishes and how its sign changes there, all invariant under Δ: the carrier scale is a gauge, not a parameter of the answer.

Contribution

A three-dimensional model, and the one-dimensional function is its chart. The layers are set out in §§2–5 and not repeated here. What the ordering buys is that each classical difficulty can be sorted into one of two boxes — a fact about the object, or a fact about the projection — and

the sorting is not a matter of taste. The abscissa of convergence, the strip, and the devices for reconstructing the functional equation from inside the projection are all in the second box; the double end, the focal cancellation and the height are in the first.

Two structures of the model are distinguished throughout because conflating them is the characteristic error. The *carrier scale* is scaling by a harmonic *constant* Δ, acting on the ruler: the sites sit at arclength $x_n = \Delta n$. The *warp* is scaling by a *function* — dynamic, unit modulus, readout-preserving — acting on the fiber; its radial generator is the log-weight operator H multiplying the n-th phasor by $\log n$, and on the carrier of scale Δ that generator is $H_\Delta = H + \log\Delta \cdot I$, the shift by a constant being the whole of the carrier's effect on it. That sentence is a theorem and not a description: H is an unbounded self-adjoint operator on $\ell^2(\mathbb{N}, \mathbb{C})$ on its maximal domain, the phasors are a Hilbert basis of its eigenvectors, $H_\Delta = H + \log\Delta \cdot I$ holds with equal domains, and the warp e^{-sH} is its functional calculus on that basis (§17). The carrier scale is a gauge — the events and the readout's sign are invariant across the family $\Delta > 0$. That invariance is not a theorem with content: Δ^{-s} cancels between the fiber and the gauge, so $u_\Delta / u_{\Delta'} = (\Delta'/\Delta)^{1/2}$, a positive constant independent of t. It is worth stating precisely because it says the model has *no free parameter to tune* — an honest design property, not a discovery. The warp is different: it genuinely moves the height at which the cancellation completes. There is no inverse warp; the reciprocal of a warp is again a warp.

It is a model for the Dirichlet L-functions, not for ζ alone. A fiber rides the carrier; the lanes are the conductor's $\pm$ channels, supplied by the character. Reality, the functional equation and the counting laws are properties of the weld involution and the readout map, and hold for every fiber that rides the carrier — ζ being the case whose formalisation is carried out here (§15 records what changes for a general primitive χ: the root number enters as a fixed ray, which the weld gauge absorbs, and nothing else does).

The chart, and what does not fit in it. The readout is the chart of the bank at a height, and in the chart it is Hardy's Z up to a strictly positive factor (Proposition 23); the isolation chain of §§7–11 is therefore what the chart reads, and it coincides with the classical argument, which is not claimed to be otherwise. What lives only in three dimensions is set out in §18: the lane structure supplies a series convergent *on* the line, which is what the locator evaluates; the same lane structure imposes a linear head, which is a cost and is reported as one; the μ_6 cell width is the constant in the refutation of Theorem 13; the lane factor supplies a closed-form off-line calibration set for the abscissa reading; the focal equivalence (Theorem 11) is a theorem about the model for the characters the Z-identification does not reach; the site weight is forced by the carrier scale being a gauge and is the climber's registration of the site (Theorem 4, §4); the distance off the line is a radial coordinate of the object, invisible to the chart (§16); and the fiber has a self-adjoint generator whose Hilbert space ends at the critical line (§17). Two of those are negative results about the carrier, which is the part we would defend hardest.

And it computes. On one side of the existing literature, geometric readings of ζ — the Dirichlet walk and its helical carrier, spirals of ζ, phasor accumulations in the Argand plane — are descriptive: they supply pictures and intuitions about where zeros can and cannot be, and no algorithm. On the other, the methods that do take an index and return an ordinate — Turing's method and its modern form, Platt's isolation, Newton refinement in ball arithmetic, Odlyzko–Schönhage — carry no geometric representation, working from the Riemann–Siegel formula and its relatives. The model here is in both columns: one addresses it by index and receives a certified interval — certified relative to the count supplied through the interface of §13, as §19(iv) spells out — and every step from the representation down to $\zeta(\frac{1}{2} + i\gamma) = 0$ is

machine-checked (§21). What is claimed is the model and its operation, not a faster algorithm and not a new localization principle; §19 is explicit about the difference.

What is proved

First, that the object and its chart are the same thing: the lane-balanced chart of the warped bank on the scaled carrier equals $\Delta^{-s}(1-2^{1-s})\zeta(s)$ on the whole channel $\operatorname{Re} s > 0$, on the three-dimensional phasors and at every carrier scale, and any entire function agreeing with that chart on the channel is $(1-2^{1-s})\zeta(s)$ everywhere (Theorem 17). The chain then runs as follows, each link machine-checked. The weld gauge makes the readout a strictly positive multiple of the completed zeta function on the line (Theorem 19), so the readout's zeros are exactly the on-line zeros of ζ, at every carrier scale (Proposition 25). Across an event of multiplicity m the readout's sign changes by $(-1)^m$, with no simplicity hypothesis (Theorem 30): multiplicity is *carried* through the argument rather than assumed away. The zeros off the line occur in conjugate pairs at a common ordinate with equal multiplicity, so the parity of the *full* strip count — not merely the on-line count — equals the readout's sign change (Theorem 34). On an interval across which the readout changes sign and on which Ξ' does not vanish, there is exactly one event and that event is simple (Theorem 38); chaining this across a ladder of ordinates gives the index (Theorem 41), and the capstone (Theorem 42) converts a sign-change sub-bracket of the i-th cell into an interval provably containing the i-th zero of ζ on the critical line. Every hypothesis of the capstone is an evaluation at finitely many points: signs of the readout at the ladder ordinates, and non-vanishing of Ξ' between them (Remark 39 says why the second is stated on Ξ and not on the readout). No zero-counting routine, no zero-density estimate, and no bound on $S(t)$ appears anywhere in the chain.

Off that chain, the operator named in §1 is proved to be one (§17): the log-weight operator is self-adjoint on ℓ^2 with the phasors as eigenbasis (Theorem 60); the warp is its functional calculus and, on the line, its unitary group (Theorem 61); ζ is its trace, $-\zeta'/\zeta$ its von Mangoldt trace, and the Euler product its decomposition into prime clocks (Theorem 62); and the fiber is its flow — the bank $B(s) = e^{-sH}\mathbb{1}_{\mathbb{R}_+}$, square-summable exactly to the right of the critical line, of which every L-series is a chart, and whose entry into the kernel of the lane-balanced chart is the chart's zero (Theorem 63).

Relation to existing work

Index to certified interval is not new, and we do not claim it. Turing's method [3] counts zeros on the line by bounding $\int_{t_1}^{t_2} S(t)\,dt$ over a suitable interval, which converts a sign count into a certified total; its modern rigorous form is due to Platt [12], who isolated the non-trivial zeros with imaginary part up to 3.061×10^{10} to absolute precision $\pm 2^{-102}$. That algorithm is implemented and shipped. Arb [9] provides `acb_dirichlet_hardy_z_zeros` and `acb_dirichlet_platt_hardy_z_zeros` take an index n and a count and return ball enclosures of consecutive zeros, the documented example retrieving zeros of index $\approx 1.048 \times 10^9$ near height 3.888×10^8 in about a quarter of a second. A reader wanting the i-th ordinate fast should use that. What is offered here is a different *representation* of the same data, with the path from the representation to $\zeta(\frac{1}{2} + i\gamma) = 0$ machine-checked.

Refinement. Once a zero is isolated, Newton's method in ball arithmetic with precision doubling refines it at the cost of roughly two full-precision evaluations at any digit count. Johansson [8] gives the rigorous high-precision evaluation machinery this rests on; the record

computation of the first ordinate to over 3×10^5 digits is a separate artifact built on that machinery and on Arb [9], not a result of [8] itself, and is recorded as the reference value in OEIS `A058303`. We make no speed claim against any of it (§19(i)).

Amortised evaluation. Odlyzko and Schönhage [10] reduce the cost of evaluating ζ at many equally spaced heights by transforming the Dirichlet sum into a rational-function evaluation at roots of unity and applying the FFT. That is an amortisation over a *region*; it does not give random access to a single index, which is what is addressed here.

The ancestor of the jet enclosure. Gargantini and Henrici [7] ask exactly the question of §12: if the value of the logarithmic derivative p'/p, and of finitely many of its derivatives, is known at a point, and the other zeros are known to lie in specified circular regions, where is the remaining zero? They answer it in circular (complex interval) arithmetic, connect it to a classical result of Laguerre, and — this is the relevant sentence — apply the theory to locating zeros of certain *transcendental* functions. The jet enclosure below is that construction with the circular regions supplied by the carrier's own tail bound, and it should be read as a descendant, not as a new localization principle. The complementary classical device, power sums of the zeros in a region by contour integration, is Delves and Lyness [1].

The Hamiltonian reading. That ζ is the partition function of a quantum statistical system whose Hamiltonian has spectrum $\{\log n\}$ is Bost and Connes [13]; the spectral side of the same frame — an operator whose eigenvalues are the ordinates — is the Hilbert–Pólya programme in the forms of Berry and Keating [14] and Connes [15]. In §17 the warp's generator of §1 is made a self-adjoint operator on ℓ^2; ζ is its trace, which is the Bost–Connes identity for this operator, and the object the trace is taken of is the *bank* — with its unitary orbit on the line, its ℓ^2-boundary at the line, and every L-series as one of its charts. No operator with the ordinates as spectrum is constructed or claimed.

2 The geometry: the double-ended helix

The object is three-dimensional, and this section fixes it before any series is written. Nothing here is a picture attached to a one-dimensional argument: the one-dimensional statements later in the paper are readouts *of* this, taken in a chart, and it is worth having the stage in hand first so that it is clear which statements are about the object and which are about the chart.

Definition 1 (the helix)**.** For a pitch $p > 0$ and a radial rate $r > 0$, the helix is the space curve

$$\gamma(k) \;=\; \big(rk\cos 2\pi k,\; rk\sin 2\pi k,\; pk\big), \qquad k \geq 0,$$

Archimedean in the turn parameter k: radius rk, pitch p. The geometry is *double-ended* — the helix together with its conjugate anti-helix, the mirror image through the origin — and both ends run $0 \to +\infty$.

LEAN: `Geometry.helix, Geometry.helixVel, Geometry.helix_hasDerivAt, Geometry.helix_cyl_radius`

The cylindrical radius of $\gamma(k)$ is $|rk|$ and the speed is

$$|\dot\gamma(k)|^2 \;=\; p^2 + r^2 + (2\pi r k)^2,$$

so the curve accelerates as it climbs: successive turns are longer, and that is the whole source of the logarithm later.

LEAN: `Geometry.helix_speed_sq, Geometry.speed`

Definition 2 (the climber, and the height)**.** The *climber* is $k(y) = e^y/p$. It registers a one-dimensional ordinate y against a point of the geometry, at cylindrical height

$$z \;=\; p\,k(y) \;=\; e^y, \qquad R(y) \;=\; r\,k(y) \;=\; (r/p)\,e^y.$$

LEAN: `Geometry.kClimb, Geometry.height_eq_p_mul_kClimb, Geometry.radius, Geometry.radius_eq, Geometry.gammaY, Geometry.gammaY_eq`

So the object is Archimedean in its own turn parameter and exponential in the analytic ordinate; both descriptions are correct, in different charts, and the climber is the change of variable between them. *The height is the geometric quantity; the ordinate is its readout* (§5). We will say that a height and an ordinate *coincide*, never that they are equal — they are quantities in different charts, related by a registration.

Theorem 3 (closed-form arclength)**.** *For $r > 0$,*

$$S(k) \;=\; \int_0^k |\dot\gamma| \;=\; \frac{k}{2}\sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2} + \frac{p^2 + r^2}{4\pi r}\,\operatorname{arsinh}\frac{2\pi r k}{\sqrt{p^2 + r^2}}.$$

For $r = 0$ it degenerates to $S(k) = pk$.

LEAN: `Geometry.arclength, Geometry.arclengthClosed, Geometry.arclength_closed_form, Geometry.arclength_r_zero`

The arclength is what the next section measures with, so it matters that it is in closed form rather than an integral to be estimated: the ruler is exact.

3 The carrier: a numberline, harmonically scaled

The geometry is the stage; it carries no marks. A *carrier* is a numberline laid along it — a ruler — and the carrier is where scaling acts. The distinction is not decorative and the paper keeps it throughout:

	scaling by	acts on
carrier scale	a harmonic *constant* Δ	the carrier
warp	a *function*, unit modulus	the fiber

Harmonic scaling does not act on the geometry. It multiplies the carrier's tick spacing by a harmonic constant Δ, and the carrier's sites then sit at equal *arclength* $x_n = \Delta n$ along the curve. The index is the arclength in units of Δ,

$$N(y) \;=\; \frac{S\big(k(y)\big)}{\Delta}, \qquad \text{and at } \Delta = \pi/3, \qquad N(y) = \frac{3}{\pi}\,S\big(k(y)\big).$$

LEAN: `Geometry.Delta, Geometry.Nindex, Geometry.Nindex_eq`

One geometry can host several carriers at once — *rails* — differently configured, running alongside one another in the same space without displacing each other. Only one rail is used in this paper.

Theorem 4 (the carrier scale forces the logarithm)**.** *Call a weight w on the carrier* gauge *if rescaling the carrier by $\Delta > 0$ shifts the weight of every site by one constant, $w(\Delta x) = w(x) + c(\Delta)$ for all $\Delta, x > 0$ — that is, if the carrier scale acts on the fiber by a factor common to all sites. Then:*

(i) $\log$ *is gauge, with* $c(\Delta) = \log \Delta$*;*

(ii) a gauge weight continuous on $(0, \infty)$ *is* $a \log x + b$ *for constants* a, b*;*

(iii) the weights of the carrier-scaled and the unscaled generator differ by the gauge constant, $\log(\Delta x_n) = \log x_n + \log \Delta$*, which is* $H_\Delta = H + \log \Delta \cdot I$ *(Theorem 60(iii)).*

LEAN: `CarrierReadout.log_isGaugeWeight, CarrierReadout.gaugeWeight_eq_log, CarrierReadout.logWeight_gauge_shift`

Proof. (i) is $\log(\Delta x) = \log x + \log \Delta$. (ii): $\varphi(u) = w(e^u) - w(1)$ satisfies $\varphi(u+v) = \varphi(u) + \varphi(v)$, by the gauge law at $\Delta = e^u$, $x = e^v$ and again at $x = 1$; a continuous additive map $\mathbb{R} \to \mathbb{R}$ is linear, so $\varphi(u) = au$ and $w(x) = a \log x + w(1)$. (iii) is (i) at the sites. □

So the site weight of the fiber is not chosen. The design fact that the carrier scale is a gauge — Proposition 25, measured in §13.3 — *is* the statement that the weight is logarithmic, by (ii); the affine freedom $a \log x + b$ is the unit of the ordinate (a) and the carrier scale ($b = \log \Delta$).

Proposition 5 (the geometric head)**.** *For* $r > 0$ *and* $k \geq 0$*,* $S(k) \geq \pi r k^2$*; hence at* $\Delta = \pi/3$ *the carrier index satisfies, for* $p, r > 0$*,*

$$N(y) \;\geq\; \frac{3r}{p^2}\, e^{2y}\,:$$

the number of sites the carrier has laid down by the ordinate y *grows exponentially in* y*, against the linear* $2y/\pi$ *that the lane saddle of §13 requires.*

LEAN: `CarrierReadout.arclength_ge_pi_r_sq, CarrierReadout.Nindex_ge_exp`

Proof. In the closed form of Theorem 3 the root is at least $2\pi r k$ and the arsinh term is non-negative, so $S(k) \geq \frac{k}{2} \cdot 2\pi r k$; then $k(y) = e^y/p$. □

That Δ is a gauge and not a parameter of the answer is the content of Proposition 25 below, and it is measured in §13.3: the extracted ordinates are identical across a factor of 57 in carrier scale.

4 The fiber: the L-function riding the carrier

The carrier is a ruler; the *fiber* is what rides it. The L-function's representation lives on the carrier, and the phasors live on the fiber. As the fiber grows from the origin outward, the conductor assigns a phasor to each carrier integer consumed — positive, negative, or neutral channels, supplied by the character. For ζ the channels are the even and odd sites and the assignment is the non-principal character mod 2; that is where the two *lanes* come from, and they are not a device introduced for ζ but what a conductor of 2 supplies.

Three properties of the fiber are used later and are worth stating as the geometry gives them, not as the series gives them.

Phasors are born at zero magnitude and grow. A phasor enters continuously at magnitude 0 at its own site and grows until finished. There is no site at the origin and no phasor before the first, and in particular there is no convergence gate: the fiber is defined at every height. (The abscissa of convergence is a fact about a projected series, §5, and it is not a constraint on the fiber.)

Phasors spin in the logarithm because the carrier turns. The site at arclength x_n carries the phasor x_n^{-s}; at $s = \frac{1}{2} + it$ this has modulus $x_n^{-1/2}$ and phase $-t \log x_n$. The logarithm is not put in by hand, and §3 has already said why once: Theorem 4 forces it. The climber of Definition 2 says it a second time, as a registration. On the axial rail — $r = 0$, where the carrier's arclength is its height (Theorem 3) — the climber reaches arclength x at the ordinate $\log x$; so it reaches the site $x_n = \Delta n$ exactly at the ordinate $\log x_n$, which is the site's weight, and at $\Delta = \pi/3$ the carrier index at that ordinate is $N(\log x_n) = n$. The phase $-t \log x_n$ is therefore the product of two registrations by the same climber: the ordinate $t = \log z$ at which the fiber is read, and the ordinate $\log x_n$ at which the site was reached. (On a rail of radial rate $r > 0$ the same site is reached at the ordinate $\log(p\, S^{-1}(x_n))$; the weight $\log x_n$ is the axial reading, and the site's *angle* — its cell — is read on the turning rail, Proposition 7.)

LEAN: `CarrierReadout.climber_reaches_arclength, CarrierReadout.climber_reaches_site_at_weight, CarrierReadout.Nindex_at_site, CarrierReadout.mellinSpin_eq_registrations, cpow_critical_line, eta_term_critical_line, norm_cpow_critical_line`

The warp. A *warp* is a scaling by a function: dynamic, unit modulus, readout-preserving. It is the geometric knob — it moves the fiber through configuration space, and turning it changes *the height at which the cancellation completes.* The warp used here aligns the fiber's vanishing with the harmonic cell edges, and the vanishing books no residue. There is no such thing as an inverse warp: the reciprocal of a warp is again a warp.

Theorem 6 (antipodal cancellation, logarithm-free)**.** *In the μ_6 cell the antipode of a cell is its negative:* $\mathrm{cell}(n) + \mathrm{cell}(n+3) = 0$.

LEAN: `Readout.cell_antipodal_cancel`

This is the mechanism in its native form: exact cancellation of the positive against the negative side, forced by the $\pi/3$ helix angle alone, with no logarithm anywhere. The vanishing of the fiber is a *focal cancellation* — the channels point at a common height and cancel there — and it is an event at a height, not a root of a formula.

Proposition 7 (the cell is the helix's angle)**.** $\mathrm{cell}(n) = e^{\,i \cdot 2\pi \cdot n/6}$*: the μ_6 cell of the site n is the angle of the helix at turn parameter $k = n/6$ — six sites per turn — and the helix at that parameter is*

$$\gamma(n/6) \;=\; \Big(r\tfrac{n}{6}\,\mathrm{Re}\,\mathrm{cell}(n),\; r\tfrac{n}{6}\,\mathrm{Im}\,\mathrm{cell}(n),\; p\tfrac{n}{6} \Big).$$

The antipode $n + 3$ of Theorem 6 is half a turn up.

LEAN: `CarrierReadout.cell_eq_helix_angle, CarrierReadout.helix_at_sixth_turn`

Definition 8 (fiber phasor, its ± block, and a cancelling partner)**.** A *fiber phasor* is a unimodular complex number w, $|w| = 1$: the state of one site of the fiber at a height. Its ± *block* is the conjugate-pair matrix $\mathrm{diag}(w, \bar{w})$ — the phasor read from the two ends of the geometry, the end-swap being conjugation (Lemma 32) — and its two entries are the phasor's *eigenheights*; they *phase-align* when $w = \bar{w}$. A *cancelling partner* of w is an $a \in \mathbb{C}$ with $w + a = 0$.

LEAN: `DeBranges.conjPairBlock`

Theorem 9 (the fiber always leaves a residue)**.** *A unimodular phasor has a unique cancelling partner, and the ± block of a unimodular fiber phasor has determinant* 1*; so the two sides cannot both vanish, and one cannot simultaneously phase-align two eigenheights and cancel exactly.*

LEAN: `Readout.fiber_residue_no_double_cancellation, Readout.cancelling_partner_unique, Readout.fiber_pair_block_det_one`

Definition 10 (the ± channels and the closure function)**.** For a real-valued character χ and an ordinate t let $a_n(t) = \frac{\pi}{3}\, n^{-1/2} e^{-it\log n}$ be the amplitude carried by the site n — the same for both channels, the character entering only through the split — and put

$$P_\chi^{(N)}(t) = \sum_{\substack{n\le N\\ \chi(n)=+1}} a_n(t), \qquad M_\chi^{(N)}(t) = \sum_{\substack{n\le N\\ \chi(n)=-1}} a_n(t),$$

the *positive* and *negative channels* truncated at N. The *closure function* at height $Z>0$ is the L-value at the corresponding ordinate, $\Phi_\chi(Z) = L(\chi, \frac{1}{2} + i\log Z)$.

LEAN: `Focal.aTerm, Focal.Pchan, Focal.Mchan, Geometric.Phi`

Theorem 11 (focal cancellation is the vanishing)**.** *For a non-principal real-valued χ and $Z>0$, the positive-minus-negative channel closes at height Z exactly when the L-value vanishes at the corresponding ordinate:*

$$P_\chi^{(N)}(\log Z) - M_\chi^{(N)}(\log Z) \xrightarrow[N\to\infty]{} 0 \quad\Longleftrightarrow\quad \Phi_\chi(Z) = 0.$$

LEAN: `Readout.focal_pm_cancellation_iff_L_zero`

This is the bridge in its native form, and it is an *iff*: the geometric event — the ± channels closing at a height — and the analytic event — the L-function vanishing — are the same occurrence seen in two charts. In the chart, Definition 10 makes $P_\chi^{(N)} - M_\chi^{(N)}$ the partial sum $\frac{\pi}{3}\sum_{n\le N}\chi(n)n^{-1/2-i\log Z}$ of the L-series at $\frac{1}{2}+i\log Z$, and the theorem's chart reading is the conditional convergence of that series on the line. The object's reading is the one displayed: the conductor sorts the sites into two channels, and the fiber vanishes at a height exactly when they close there.

Remark 12 (ζ is not covered by Theorem 11, and what covers it instead). The hypothesis $\chi \neq 1$ excludes the trivial character, so Theorem 11 does *not* apply to ζ. This is not an oversight of the statement: the ± split there is by character *value*, and for ζ that split degenerates — there is no non-trivial sign to sort by. Saying otherwise would be the characteristic error of this subject, and we do not.

For ζ the corresponding decomposition is by *angle* rather than by value: the bank's phasors, with von Mangoldt weights and no character hypothesis, are sorted into the six μ_6 sectors by $t\log n$, and antipodal sectors are compared. The decomposition is an exact partition, the antipodal pairing is a fixed-point-free involution, and the imbalance vanishes at a truncation exactly when all three antipodal pairs agree — all machine-checked.

LEAN: `ZetaSectorBank.sector_partition, ZetaSectorBank.bank_pair_decomposition, ZetaSectorBank.antipode_involutive, ZetaSectorBank.imbalance_eq_zero_iff, ZetaSectorBank.norm_zetaTerm`

The balance law built on top of that bookkeeping — that the sectors balance in the limit at every on-line zero — was carried in an earlier version of this work as a named hypothesis. It is not open. It is false, and the reason is elementary.

Theorem 13 (the antipodal arc obstruction)**.** *For every real t and every N,*

$$\frac{\sqrt{3}}{2}\sum_{n\le N}\frac{\Lambda(n)}{\sqrt{n}} \;\le\; \mathrm{imb}(N,t), \qquad \mathrm{imb}(N,t) = \sum_{j=0}^{2}\big|S_j(N,t) - S_{j+3}(N,t)\big|.$$

Consequently $\mathrm{imb}(N,t) \not\to 0$ *for* every *real* t*: the balance law fails at every on-line zero, and since* ζ *has on-line zeros (*$\gamma_1 = 14.134\ldots$*, certified in §16) the law is false. The compiled form of that last step is an equivalence, the law holding if and only if* ζ *has no zero on the critical line at all.*

LEAN: `ZetaSectorBank.imbalance_ge_half_mass, ZetaSectorBank.not_sectorBalanceAt, ZetaSectorBank.angleSectorBalanceLaw_iff_no_critical_zero, ZetaSectorBank.re_rotate_ge, ZetaSectorBank.re_rotate_le, ZetaSectorBank.norm_sub_ge, ZetaSectorBank.floorConst_le_imbalance`

Proof. A sector is an arc of angular width $2\pi/6 = \pi/3$. Every phasor in sector j therefore lies within $\pi/6$ of that sector's midpoint direction u_j, so its component along u_j is at least $\cos(\pi/6) = \sqrt{3}/2$ times its modulus. The antipodal sector $j+3$ is the same arc rotated by π, so its phasors have midpoint direction $-u_j$ and their components along u_j are at most $-\sqrt{3}/2$ times their moduli. Subtracting therefore *adds* the two masses:

$$|S_j - S_{j+3}| \;\geq\; \langle S_j - S_{j+3},\, u_j\rangle \;\geq\; \frac{\sqrt{3}}{2}\big(M_j + M_{j+3}\big),$$

where M_j is the mass in sector j; and the three antipodal pairs sweep the six sectors exactly once, giving the displayed bound. For the consequence, no asymptotics are needed: the $n = 2$ term alone gives $\mathrm{imb}(N,t) \geq \frac{\sqrt{3}}{2} \cdot \frac{\log 2}{\sqrt{2}} > 0.42$ for every $N \geq 2$, and a sequence bounded below by a positive constant does not tend to 0. □

Measured, for confirmation rather than for evidence: at $\gamma_1 = 14.1347\ldots$ and at the non-zero $t = 20$, the ratio imb /mass sits at 0.955 for every N from 10^3 to 4×10^6 — above the proved floor $\sqrt{3}/2 = 0.866$, and *identical at the zero and the non-zero to three decimals*. So the statistic does not merely fail to vanish; it does not distinguish zeros from non-zeros at all. Two independent implementations agree — one in `numpy` with a sieve, one in `mpmath` at 40 digits with trial division and no `numpy` at all (`checks/balance_law_refuted.py`, `checks/balance_law_independent.py`); they match to six digits.

Remark 14. This is reported because it removes the one place where this work carried an unproved hypothesis, and it removes it in the direction that is easy to get wrong. The correct conclusion is not that the μ_6 geometry is idle — the log-free antipodal cancellation of Theorem 6 is exact and is the mechanism — but that *this particular statistic*, sorting the von Mangoldt bank into six arcs and differencing antipodes, cannot detect a zero, because differencing an arc against its own antipode is a sum and not a cancellation. Nothing elsewhere in the paper used it.

So the licence for the chart work on ζ is not Theorem 11 and not the balance law. It is an exact algebraic identity, and that is the stronger footing: the fiber *factors* on the line as $F_\Delta(\frac{1}{2} + it) = w_\Delta(t)\,\Lambda(\frac{1}{2} + it)$ with $w_\Delta(t) \neq 0$ (Theorem 19), so the readout's vanishing and ζ's vanishing are the same set — not in a limit, not conditionally, but by a factorisation. Everything from §6 onward rests on that identity. Theorem 11 and Remark 12 say what the same phenomenon looks like in its geometric form, and how much of that form is currently proved.

5 The descent to the chart

The one-dimensional L-function is not the object; it is the object read in a chart, and the chart is reached by two projections, each with its own loss.

step	map	drops	survives
3-D $\to$ 2-D	Cayley/Möbius	the radius	angle, height
2-D $\to$ 1-D	$y = \log z$	the angle	the height

Theorem 15 (the two steps)**.** *The Cayley map $w \mapsto (w-i)/(w+i)$ carries the real readout line to the unit circle, $\|\mathrm{cayley}(x)\| = 1$ for real x; and the height readout $y = \log z$ inverts globally, $\log(e^{\gamma}) = \gamma$, with no branch cut.*

LEAN: `Readout.cayley, Readout.cayley_real_norm_one, Readout.heightReadout, Readout.readout_exp, Readout.height_eq_exp_of_readout`

Two consequences are used repeatedly and are stated here so that they are not rediscovered later as difficulties.

The classical strip is the Cayley image of the unit circle. It is a feature of the chart, not of the geometry, and phenomena that live in it are chart artifacts. The geometry has no free abscissa to vary: there is a height axis, and the critical abscissa is not posited there but is where the double-ended reflection has its fixed set. In particular the abscissa of convergence $\mathrm{Re}\, s > 1$ is where a projected series stops converging; the fiber has no convergence gate, and this paper never treats the strip's boundary as a constraint on the object.

The height survives; the ordinate is its registration. The logarithm appears exactly once in the whole model, in the 2-D $\to$ 1-D step, and that is the only place it appears. A vanishing event happens at a height z; its readout is the ordinate $\gamma = \log z$. The two coincide; they are not the same quantity.

The functional equation is the double end. The geometry has two ends, mirror images through the origin, and a readout cannot tell which end it is reading. That end-swap descends through the Cayley step to $s \mapsto 1 - s$. The functional equation is therefore not an identity to be derived in the chart but a symmetry the object simply has; the chart's devices for reconstructing it from inside the projection are chart devices, and the paper does not use them to explain carrier facts.

6 The chart readout: lanes and the weld gauge

From here to §13 the work is in the chart, and every object is a readout of §§2–4. The reason for descending is Theorem 11: the events are the same, and in the chart they are computable and machine-checkable.

Definition 16 (carrier fiber)**.** For $\Delta > 0$ set

$$F_{\Delta}(s) \;=\; \Delta^{-s}\big(1 - 2^{1-s}\big)\zeta(s),$$

the sum over the sites $x_n = \Delta n$ of the phasors x_n^{-s} with the odd and even sites entering with opposite sign.

LEAN: `carrierFiber`

Theorem 17 (the fiber is the function)**.** *Let $\Delta > 0$; write $B_{\Delta}(s) = (x_n^{-s})_{n \ge 1}$ for the bank on the carrier of scale Δ, $x_n = \Delta n$, and ℓ for the lane-balanced chart, the phasors summed in pairs in the order they enter.*

(i) $B_\Delta(s) = e^{-sH_\Delta\not\Vdash}$*: the phasor at the site* x_n *is* $\exp(-s\,\lambda_n(H_\Delta))$ *with* $\lambda_n(H_\Delta) = \log x_n$ *the eigenvalue of the scaled generator (§17); and* $B_\Delta(s) = \Delta^{-s}B_1(s)$*, the carrier scale a gauge on the bank.*

(ii) For $\operatorname{Re} s > 0$*,* $s \neq 1$*:* $\ell(B_\Delta(s)) = F_\Delta(s)$*. The chart of the warped bank on the scaled carrier* is *the fiber —* $\Delta^{-s}(1-2^{1-s})\zeta(s)$*, the analytic function and not a truncation of it — on the whole channel on which the ordered sum converges.*

(iii) In the strip, $\zeta(s) = 0$ *if and only if* $B_\Delta(s) \in \ker \ell$*, at every carrier scale.*

(iv) The same identity holds on the three-dimensional phasors: with χ_2 *the conductor-2 lane character, the planar projections of the 3-D phasors* $\big(\chi_2(n)\,n^{-\sigma}e^{-iy\log n},\ (1-|\chi_2(n)|)\,n^{-\sigma}\big)$*, summed in pairs in the order they enter and scaled by* Δ^{-s}*, equal* $F_\Delta(\sigma+iy)$ *for* $\sigma > 0$*,* $\sigma + iy \neq 1$*. (The mass-axis coordinate vanishes: for* ζ *the neutral channel is empty.)*

(v) The one-dimensional function has no content beyond the chart: any entire function that agrees with $\ell(B_1(s))$ *on the channel equals* $(1-2^{1-s})\zeta(s)$ *at every* $s \neq 1$*.*

LEAN: `CarrierReadout.fiber_eq_function, CarrierReadout.fiber3D_eq_function, CarrierReadout.function_determined_by_fiber, CarrierReadout.plane_phasor3D_eq, CarrierReadout.laneChart3D_eq, CarrierReadout.bank_eq_rpow_mul_spin, DVP.etaLim_eq_zeta`

Proof sketch. (i): $x_n^{-s} = e^{-s\log x_n}$, and $(\Delta n)^{-s} = \Delta^{-s}n^{-s}$ (`CarrierReadout.bankΔ_eq_exp`, `CarrierReadout.bankΔ_eq_scale`). (ii): pairing consecutive phasors gives the ordered sum $\sum_k\big((2k+1)^{-s} - (2k+2)^{-s}\big)$, which converges for $\operatorname{Re} s > 0$ and equals $(1-2^{1-s})\zeta(s)$ there — the continuation of the lane-balanced series across the abscissa of the unbalanced one, proved in the sources from $\operatorname{Re} s > 1$ by the identity theorem (`DVP.etaLim_eq_zeta`); then the gauge of (i) (`CarrierReadout.laneChart_bankΔ`). (iii): the lane factor does not vanish in the strip (`CarrierReadout.zeta_zero_iff_bankΔ_mem_ker`). (iv): the planar projection of the 3-D phasor at site n is $\chi_2(n)\,n^{-\sigma}e^{-iy\log n} = \chi_2(n)\,n^{-s}$. (v): both sides are analytic on $\mathbb{C}\setminus\{1\}$, which is connected, and agree on the open channel. □

Before this theorem the closed form of Definition 16 was, in the sources, the *definition* of `carrierFiber`; the description "the sum over the sites of the phasors, the odd and even sites entering with opposite sign" is now what is proved, on the three-dimensional phasors and at every carrier scale. The boundary is stated exactly: the ordered sum converges on $\operatorname{Re} s > 0$ and nowhere to the left of it, and there the function is determined by the chart, which is (v).
The lane-balanced sum converges for $\operatorname{Re} s > 0$ — that is the whole point of taking the lanes with opposite sign, and it is why the construction can be written *on* the critical line rather than continued to it. The unbalanced Dirichlet series would need $\operatorname{Re} s > 1$; the alternating one does not, and the displayed closed form is an identity of convergent objects at $s = \frac{1}{2} + it$. The convergence is conditional, which is exactly the fact the head law of §13 has to pay for.

Definition 18 (weld factor)**.**

$$w_\Delta(t) \;=\; \frac{\Delta^{-(\frac{1}{2}+it)}\big(1-2^{\frac{1}{2}-it}\big)}{\Gamma_{\mathbb{R}}(\frac{1}{2}+it)}, \qquad \Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2).$$

LEAN: `weldFactor`

Theorem 19 (the weld gauge)**.** *For every $\Delta > 0$ and every $t \in \mathbb{R}$,*

$$e^{-i \arg w_\Delta(t)} F_\Delta(\tfrac{1}{2} + it) \;=\; |w_\Delta(t)|\, \Lambda(\tfrac{1}{2} + it) \;\in\; \mathbb{R},$$

and $|w_\Delta(t)| = \Delta^{-1/2} |1 - 2^{\frac{1}{2} - it}| / |\Gamma_{\mathbb{R}}(\frac{1}{2} + it)| > 0$.

LEAN: `weld_gauge_exact, weld_gauge_im_zero, weldFactor_norm, carrierFiber_eq_weldFactor_mul, lane_factor_ne_zero, weldFactor_ne_zero`

Proof sketch. The factorisation $F_\Delta(\frac{1}{2} + it) = w_\Delta(t)\Lambda(\frac{1}{2} + it)$ is $\eta = (1 - 2^{1-s})\zeta$ together with $\Lambda = \Gamma_{\mathbb{R}}\zeta$, the Δ^{-s} cancelling between fiber and gauge. Rotating a nonzero complex number by its own argument lands it on its modulus, so the left side is $|w_\Delta(t)|\, \Lambda(\frac{1}{2} + it)$; and $\Lambda(\frac{1}{2} + it) \in \mathbb{R}$ by the functional equation together with Schwarz reflection (§21). Positivity of $|w_\Delta(t)|$ is the observation that $1 - 2^{\frac{1}{2} - it}$ never vanishes, since $|2^{\frac{1}{2} - it}| = \sqrt{2} \neq 1$; and that $\Gamma_{\mathbb{R}}(\frac{1}{2} + it)$ is finite and nonzero, since Γ has no zeros and $\frac{1}{4} + \frac{it}{2}$ is never a pole. □

Remark 20. No asymptotics enter. The Riemann–Siegel theta function never appears; its role is played by $\arg \Gamma_{\mathbb{R}}(\frac{1}{2} + it)$, which it equals. Indeed $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ gives

$$\arg \Gamma_{\mathbb{R}}(\tfrac{1}{2} + it) \;=\; \arg \Gamma(\tfrac{1}{4} + \tfrac{it}{2}) - \tfrac{t}{2} \log \pi \;=\; \vartheta(t),$$

which is the definition of ϑ. The equality is one of *arguments*, hence modulo 2π: arg is the principal value and ϑ the continuous branch, and the two differ by the winding $2\pi k(t)$. Nothing tracks that branch, because w_Δ uses the argument only through $e^{-i \arg \Gamma_{\mathbb{R}}}$, on which k acts trivially. This is the whole reason the construction needs no asymptotic expansion of ϑ: the classical device computes ϑ to subtract it, and here it is divided out exactly by the local factor it came from.

The script `extractor/checks/theta_identity.py` checks this numerically at 60-digit working precision, at thirteen sample heights spanning $t \in [0, 10^4]$, including $t = 2\pi$ and the first ordinate: worst branch-corrected discrepancy 2.5×10^{-57}, i.e. agreement to working precision at every sample. The check was pre-registered to refute the remark at any sample exceeding 10^{-30}.

Definition 21 (the readout)**.** $u_\Delta(t) = |w_\Delta(t)|\, \mathrm{Re}\, \Lambda(\frac{1}{2} + it)$.

LEAN: `weldReadout, xiRe`

Definition 22 (event)**.** An *event* is a real t at which the readout vanishes — equivalently, by Proposition 25 below, a zero of ζ on the critical line. Its *multiplicity* is the order of vanishing there, and an event of multiplicity 1 is called *simple*. Counts of events are always counts with multiplicity unless said otherwise. The word is used rather than "zero" as a standing reminder that an event is an *on-line* zero: the readout sees exactly these, and whether they exhaust the zeros of ζ in a band is the question §8 keeps open and §19(iv) restates.

Proposition 23 (the readout is Hardy's Z in the carrier's gauge)**.** *For every $\Delta > 0$ and every real t,*

$$u_\Delta(t) \;=\; |w_\Delta(t)|\, \big|\Gamma_{\mathbb{R}}(\tfrac{1}{2} + it)\big|\, Z(t), \qquad Z(t) = e^{i\vartheta(t)} \zeta(\tfrac{1}{2} + it),$$

and both factors in front are strictly positive. Consequently u_Δ and Z have the same zeros, the same orders and the same sign at every t.

Proof. $\Lambda(\frac{1}{2}+it) = \Gamma_{\mathbb{R}}(\frac{1}{2}+it)\zeta(\frac{1}{2}+it)$ and $\vartheta = \arg\Gamma_{\mathbb{R}}(\frac{1}{2}+it)$ (Remark 20), so $\Lambda(\frac{1}{2}+it) = |\Gamma_{\mathbb{R}}(\frac{1}{2}+it)|\,e^{i\vartheta(t)}\zeta(\frac{1}{2}+it) = |\Gamma_{\mathbb{R}}(\frac{1}{2}+it)|\,Z(t)$. Multiply by $|w_\Delta(t)|$ and apply Definition 21. Positivity of the two factors is Theorem 19. □

Remark 24 (what this costs and what it buys). It should be said plainly, because it governs how the rest of the paper is to be read. *The carrier readout is Hardy's Z, up to a strictly positive factor.* It follows that §§7–11 — reality on the line, the sign change across an odd-order zero, uniqueness and simplicity by Rolle and the intermediate value theorem — are the classical Z/Ξ isolation argument. We do not claim them as new mathematics, and a reader who already knows that argument will recognise every step.

What the identification buys is the other direction, and it is used twice. First, the certification is *evaluator-agnostic*: every theorem here consumes u_Δ only through its sign, so the sign may be taken from any evaluator of Z — which is what closes (O4), and it is why a two-ended head can be substituted for the lane bank without touching a proof. Second, it is what makes the geometry's contribution locatable rather than decorative: since the isolation argument is classical, the question "what does the carrier actually do here?" has a definite answer, and §18 gives it.

Proposition 25 (events)**.** $u_\Delta(t) = 0 \iff \zeta(\frac{1}{2}+it) = 0$*, for every* $\Delta > 0$*; and the sign of* u_Δ *does not depend on* Δ*.*

LEAN: `weldReadout_eq_zero_iff, weldReadout_carrier_ratio, weldReadout_sign_carrier_free`

Proof sketch. By Theorem 19 the readout is $|w_\Delta(t)|\Lambda(\frac{1}{2}+it)$ with $|w_\Delta(t)| > 0$, so it vanishes exactly where $\Lambda(\frac{1}{2}+it)$ does; and $\Lambda = \Gamma_{\mathbb{R}}\zeta$ with $\Gamma_{\mathbb{R}}(\frac{1}{2}+it)$ finite and nonzero, so that is exactly where $\zeta(\frac{1}{2}+it)$ does. Changing Δ multiplies the readout by the positive ratio $|w_\Delta(t)|/|w_{\Delta'}(t)| = (\Delta'/\Delta)^{1/2}$, which cannot change a sign. □

7 The ordinate variable and the entire readout

Everything so far has been written on the line $s = \frac{1}{2}+it$, where the readout is a function of the real variable t. For the counting arguments we need an *entire* function of a complex variable whose restriction to the real axis is that readout, so that order, multiplicity and the argument principle are available. The substitution that does this is $s = \frac{1}{2}+iz$: the functional equation's involution $s \mapsto 1-s$ becomes $z \mapsto -z$, and complex conjugation on the line becomes conjugation in z. This is Riemann's own change of variable [2], not something the carrier supplies, and we record it as his.

Write $\Lambda_0 = \texttt{completedRiemannZeta}_0$ for the entire completed zeta, so that $\Lambda(s) = \Lambda_0(s) - \frac{1}{s} - \frac{1}{1-s}$.

Definition 26 (the entire readout)**.**

$$\Xi(z) \;=\; \big(\tfrac{1}{2}+iz\big)\big(\tfrac{1}{2}-iz\big)\,\Lambda_0\big(\tfrac{1}{2}+iz\big) \;-\; 1.$$

LEAN: `xiEntire, xiEntire_differentiable`

Remark 27 (normalisation). This Ξ is not Riemann's. With $\xi(s) = \frac{1}{2}s(s-1)\Lambda(s)$ one has $s(1-s)\Lambda(s) = -2\xi(s)$, so

$$\Xi(z) \;=\; -2\,\xi(\tfrac{1}{2}+iz),$$

an overall factor of -2 against the classical normalisation. Nothing below depends on the constant — every statement is about zeros, orders and sign *changes* — but a reader comparing formulas against Riemann's or Titchmarsh's should carry it.

Entirety is manifest: Λ_0 is entire and the remaining factors are polynomials. No pole is cancelled by hand and no removable singularity is argued away — the -1 is exactly what the two poles contribute, as the next statement records.

Proposition 28 (pole clearing)**.** *For z with $\frac{1}{2}+iz \notin \{0,1\}$, $\Xi(z) = (\frac{1}{2}+iz)(\frac{1}{2}-iz)\,\Lambda(\frac{1}{2}+iz)$; that is, Ξ is $s(1-s)\Lambda(s)$ in the ordinate variable. On the real axis,*

$$\Xi(t) \;=\; \big(\tfrac{1}{4}+t^2\big)\,\Lambda\big(\tfrac{1}{2}+it\big), \qquad t \in \mathbb{R},$$

and $\frac{1}{4}+t^2 > 0$, so $\Xi \restriction \mathbb{R}$ is a strictly positive multiple of the readout: same sign, same zeros, same orders.

LEAN: `xiEntire_eq_completed, xiEntire_ofReal, quarter_add_sq_pos`

Proof sketch. Write $A = \frac{1}{2}+iz$ and $B = \frac{1}{2}-iz$, so that $A+B=1$. With $\Lambda = \Lambda_0 - \frac{1}{A} - \frac{1}{B}$ at $s=A$, the clearing is the identity

$$AB\Big(L - \tfrac{1}{A} - \tfrac{1}{B}\Big) \;=\; ABL - 1,$$

valid for any L whenever $A, B \neq 0$ and $A+B=1$: expand the left side to $ABL - B - A$ and substitute $A+B=1$. The two factors are never inspected — only their sum is used. On the real axis $AB = (\frac{1}{2}+it)(\frac{1}{2}-it) = \frac{1}{4}+t^2$, which is positive, giving the displayed form. □

Proposition 29 (reality)**.** $\Xi(t) \in \mathbb{R}$ *for every real t, and* $\Xi(t) = 0 \iff \zeta(\frac{1}{2}+it) = 0$.

LEAN: `xiEntire_im_ofReal, xiEntire_ofReal_eq_zero_iff, xiEntire_re_eq_zero_iff`

Proof sketch. Immediate from Proposition 28: $\Xi(t)$ is the real number $\frac{1}{4}+t^2$ times $\Lambda(\frac{1}{2}+it)$, which is real, and the multiplier is positive, so $\Xi(t)=0$ exactly when $\Lambda(\frac{1}{2}+it)=0$, i.e. exactly when $\zeta(\frac{1}{2}+it)=0$. □

So Ξ is a real-on-$\mathbb{R}$ entire function with the on-line zeros of ζ as its real zeros, carrying the same multiplicities. Every count below is a count of zeros of Ξ.

Theorem 30 (multiplicity sign law)**.** *Let G be analytic at a real point t_0, real on the real axis, not identically zero near t_0, of order m there. Then for all sufficiently small $h > 0$,*

$$(-1)^m\ \mathrm{Re}\,G(t_0-h)\ \mathrm{Re}\,G(t_0+h)\ >\ 0.$$

Consequently the carrier readout's sign is multiplied by $(-1)^m$ across an event of multiplicity m, at every carrier scale.

LEAN: `sign_flip_of_order, weldReadout_sign_flip`

Proof sketch. Factor $G(z) = (z-t_0)^m g(z)$ with g analytic and $g(t_0) \neq 0$. Along the real axis $(t-t_0)^m$ is a nonzero real for $t \neq t_0$, so g inherits reality there by a punctured limit, and g keeps the sign of $g(t_0)$ on a small enough interval. Then

$$\mathrm{Re}\,G(t_0-h)\ \mathrm{Re}\,G(t_0+h) \;=\; (-h)^m h^m\, g(t_0-h)g(t_0+h) \;=\; (-1)^m h^{2m}\, g(t_0-h)g(t_0+h),$$

and the last product is positive for small h. Multiplying by $(-1)^m$ gives the claim. The carrier statement follows because the readout is a positive multiple of $\Xi \restriction \mathbb{R}$ (Remark 39). □

Remark 31. The statement is with multiplicity. This is what removes any simplicity hypothesis from the chain; it is also the honest price of a sign bit, which cannot see an event of even order.

8 Off-line pairing and the strip count

Lemma 32 (the reflection is conjugation)**.** *In the ordinate variable the involution $s \mapsto 1 - \bar{s}$, whose fixed set is the critical line, is $z \mapsto \bar{z}$.*

LEAN: `crit_is_conj, ordinate_eq, offCriticalLine_iff`

Proof sketch. With $s = \frac{1}{2} + iz$ one has $\bar{s} = \frac{1}{2} + \overline{iz} = \frac{1}{2} - i\bar{z}$, hence

$$1 - \bar{s} \;=\; 1 - \tfrac{1}{2} + i\bar{z} \;=\; \tfrac{1}{2} + i\bar{z},$$

which is the point of ordinate $\bar{z}$. So $s \mapsto 1 - \bar{s}$ reads as $z \mapsto \bar{z}$, whose fixed set is $\operatorname{Im} z = 0$, i.e. $\operatorname{Re} s = \frac{1}{2}$. □

Theorem 33 (off-line pairing)**.** *A zero of Ξ off the real axis — a zero of ζ off the critical line, which is not an event in the sense of Definition 22 — has a distinct partner at the same ordinate with the same multiplicity.*

LEAN: `xiEntire_conj, xiEntire_order_conj, offAxis_pairs, offCriticalLine_partner`

Proof sketch. The key identity is $\Xi(\bar{z}) = \overline{\Xi(z)}$, and it consumes *both* the conjugation symmetry $\overline{\Lambda_0(\bar{s})} = \Lambda_0(s)$ and the functional equation $\Lambda_0(1 - s) = \Lambda_0(s)$; neither alone suffices, because $z \mapsto \bar{z}$ is the composite $s \mapsto 1 - \bar{s}$ of the two. Given it, $\Xi(z) = 0$ forces $\Xi(\bar{z}) = 0$, and $\bar{z} \neq z$ precisely when z is off the axis, which by Lemma 32 is off the critical line. Orders agree by transporting the local factorisation $\Xi(w) = (w - z)^m h(w)$ through conjugation. □

Theorem 34 (the strip count's parity)**.** *For a bounded height band with no event at either endpoint, the parity of the total multiplicity of* all *events in the band — on the critical line or off it — is the readout's sign change.*

LEAN: `bounded_strip_zeros_finite, even_sum_of_involution, offline_strip_multiplicity_even, full_count_parity_eq_weld_parity`

Proof sketch. The band is a compact rectangle in the ordinate variable and Ξ is entire and not identically zero, so it holds finitely many events. Off the axis, conjugation is a fixed-point-free involution preserving the order, so the off-line zeros split into two-element orbits, each contributing $2m$ to the total — an even number, whatever m is. The total multiplicity is therefore congruent mod 2 to the on-line total, and the on-line half is Theorem 36. □

Theorem 36 is stated and proved in §9, and its proof uses nothing from this section, so the forward reference carries no circularity. That is mechanically checkable and was checked: none of the twelve identifiers defined in `CarrierOffLinePairing.lean`, the module carrying this section, occurs anywhere in `CarrierIntervalCount.lean`, the module carrying §9.

Proposition 35 (the boundary)**.** *Every event in the band is interior:* $0 < \operatorname{Re} s < 1$.

LEAN: `xiEntire_ne_zero_of_im_eq_neg_half, xiEntire_ne_zero_of_im_eq_half, stripBand_zeros_interior, stripBand_zeros_in_open_strip`

Proof sketch. In the ordinate variable $\operatorname{Re} s = \frac{1}{2} - \operatorname{Im} z$, so the two edges $\operatorname{Re} s \in \{0, 1\}$ are $\operatorname{Im} z = \pm\frac{1}{2}$. On the edge $\operatorname{Re} s = 1$ non-vanishing is $\zeta(1 + it) \neq 0$; the edge $\operatorname{Re} s = 0$ follows from it by the functional equation. The one point where Λ has a pole rather than a value is handled by the entire form: at $z = -i/2$, i.e. $s = 1$, the factor $\frac{1}{2} - iz$ vanishes and $\Xi(-i/2) = 0 \cdot \Lambda_0(1) - 1 = -1 \neq 0$. □

9 The interval count

Theorem 36 (interval count parity)**.** *For $a \leq b$ with no event at either endpoint,*

$$\operatorname{ord}(a, b) \text{ is even} \iff u_\Delta(a)\, u_\Delta(b) > 0,$$

where $\operatorname{ord}(a, b)$ *is the total multiplicity of the on-line events in* (a, b)*.*

LEAN: `zeros_finite, sign_const_of_no_zeros, zeroCount_split, interval_sign_parity, zeroCount_even_iff, weldReadout_interval_parity`

Proof sketch. Strong induction on the number of events in (a, b), which is finite. With none, the readout is continuous and non-vanishing on $[a, b]$, hence of constant sign, so the product is positive and the count is 0. Otherwise isolate one event t_0, split (a, b) at points either side of it at which the readout does not vanish, apply Theorem 30 across t_0 to pick up a factor $(-1)^m$, and apply the inductive hypothesis to the two shorter intervals; the counts add and the signs multiply. □

Theorem 37 (scope)**.** *The on-line count equals the full strip count for a band exactly when no event in that band lies off the critical line.*

LEAN: `stripCount_eq, stripCount_eq_zeroCount_iff`

This is stated so that the index certified by the on-line chain is not confused with the classical index. The machinery nowhere assumes the two agree; where the distinction matters it is flagged, in §19(iv).

10 A count of one without a count

Theorem 38. *Let $a < b$. If Ξ' does not vanish on $[a, b]$ and $u_\Delta(a)u_\Delta(b) < 0$, then (a, b) contains exactly one event and it is simple.*

LEAN: `zeroCount_eq_one_of_deriv_ne_zero, subsingleton_zeros_of_deriv_ne_zero, exists_zero_of_sign_change, order_eq_one_of_deriv_ne_zero`

Proof sketch. Three steps. *At least one:* the readout is continuous and changes sign, so the intermediate value theorem supplies an event. *At most one:* if there were two, Rolle applied to $\Xi \restriction \mathbb{R}$ between them would give a point of $[a, b]$ with $\Xi' = 0$, contrary to hypothesis. *Simple:* at the event γ we have $\Xi(\gamma) = 0$ and $\Xi'(\gamma) \neq 0$, which is order exactly one. □

Remark 39 (which function's derivative, and why it matters). The sign hypothesis is the carrier readout's and the derivative hypothesis is Ξ's, and the two are not interchangeable. By Proposition 28 and Theorem 19,

$$u_\Delta(t) \;=\; \rho_\Delta(t)\, \Xi(t), \qquad \rho_\Delta(t) \;=\; \frac{|w_\Delta(t)|}{\frac{1}{4} + t^2} \;>\; 0,$$

so u_Δ and Ξ have the same zeros and the same sign, and the sign hypotheses transfer exactly, at every carrier scale — that is the content of `sign_product_iff`. Their derivatives, however, satisfy $u'_\Delta = \rho'_\Delta\,\Xi + \rho_\Delta\,\Xi'$, and away from an event the first term need not vanish; so $u'_\Delta \neq 0$ on $[a,b]$ and $\Xi' \neq 0$ on $[a,b]$ are genuinely different hypotheses, and it is the latter that is formalised.

At an event the distinction disappears: $\Xi(\gamma) = 0$ forces $u'_\Delta(\gamma) = \rho_\Delta(\gamma)\,\Xi'(\gamma)$ with $\rho_\Delta(\gamma) > 0$, so $u'_\Delta(\gamma) \neq 0 \iff \Xi'(\gamma) \neq 0$ and "simple" means the same thing read either way. What is *not* formalised here is the version of Theorem 38 whose derivative hypothesis is $u'_\Delta \neq 0$; it is true, by the same Rolle argument applied to u_Δ, but it is not what the Lean statements consume, and we state the hypothesis as it is checked.

Remark 40 (why not a density bound). The apparent alternative — "a short interval holds at most two events" — is not a weaker statement. Recall the Riemann–von Mangoldt decomposition of the counting function $N(T)$ of zeros with ordinate in $(0,T]$,

$$N(T) \;=\; \frac{\vartheta(T)}{\pi} + 1 + S(T), \qquad S(T) \;=\; \frac{1}{\pi}\arg\zeta(\tfrac{1}{2} + iT),$$

the argument taken by continuous variation; S is the entire difficulty, and it is what every classical count has to control. Since $N(b) - N(a) = \big(\vartheta(b) - \vartheta(a)\big)/\pi + S(b) - S(a)$ and the main term is $w\log(t/2\pi)/2\pi + O(w^2)$ for a bracket of width $w = b - a$, bounding the count by two amounts to

$$S(b) - S(a) \;\le\; 2 - \frac{w\log(t/2\pi)}{2\pi} + O(w^2).$$

So the "density bound" is a bound on the variation of S over a short interval — the same quantity, not an easier one. And approaching it from the other side does not help either: Jensen's formula on a disc of radius comparable to the spacing gives $O(\log t)$ events, not two. The hypothesis of Theorem 38 is pointwise and computable instead, which is the whole reason it is the one used.

11 The chain and the capstone

Theorem 41 (the index from a certified chain)**.** *Let $t_0 < t_1 < \cdots < t_n$ be such that across each cell the readout changes sign and Ξ' does not vanish. Then* $\mathrm{ord}(t_0, t_n) = n$*; and for $1 \le i \le n$ the counts to either side of the i-th cell are $i-1$ and i, and that cell holds a unique event, which is simple.*

LEAN: `zeroCount_chain, ith_event_of_chain`

Proof sketch. Theorem 38 applied to each cell gives exactly one event there, simple, so each cell contributes 1; the counts add over the ladder by `zeroCount_split`, giving $\mathrm{ord}(t_0, t_n) = n$ and $\mathrm{ord}(t_0, t_{i-1}) = i - 1$, $\mathrm{ord}(t_0, t_i) = i$. □

Theorem 42 (the capstone)**.** *Under the hypotheses of Theorem 41, for any $t_{i-1} \le c \le d \le t_i$ with $u_\Delta(c)u_\Delta(d) < 0$ there is $\gamma \in (c,d)$ with $\zeta(\frac{1}{2} + i\gamma) = 0$, and γ is the unique event of the i-th cell.*

LEAN: `ith_event_enclosed, sign_product_iff, ith_event_enclosed_carrier`

Proof sketch. By Theorem 41 the i-th cell (t_{i-1}, t_i) holds a unique event γ. The sub-bracket $[c,d]$ lies inside that cell and the readout changes sign across it, so by the intermediate value theorem it contains an event; uniqueness in the cell forces that event to be γ. Proposition 25 converts "event" to $\zeta(\frac{1}{2} + i\gamma) = 0$. □

Every hypothesis is a finite computation: $n+3$ sign evaluations of the readout, and non-vanishing of Ξ' on each cell (Remark 39).

12 The jet enclosure

Theorem 42 encloses the i-th event in a bracket (c, d) obtained from sign evaluations alone. That bracket is as narrow as the bisection that produced it and no narrower. This section sharpens it to an interval of width 2ε using the logarithmic jets of Ξ at a nearby base point.

Fix a base point y_0 and set $d = y_0 - \gamma$ for the event γ in question. The jets are the power sums

$$A_m \;=\; \sum_j (y_0 - \gamma_j)^{-m},$$

and the classical observation is that $A_m/A_{m+1} \to d$ when γ is the ordinate nearest y_0. What we need is not the limit but an *enclosure* valid at finite m, with every error that entered the computation carried explicitly.

Theorem 43 (the ratio enclosure)**.** *Let $d \in \mathbb{C}^\times$, $m \in \mathbb{N}$, and let $P, Q \in \mathbb{C}$ and $\alpha, \beta \geq 0$ satisfy*

$$\big|P - d^{-m}\big| \leq \alpha, \qquad \big|Q - d^{-(m+1)}\big| \leq \beta, \qquad \beta < |d|^{-(m+1)}.$$

Then

$$\Big|\frac{P}{Q} - d\Big| \;\leq\; \frac{\alpha + |d|\,\beta}{|d|^{-(m+1)} - \beta}.$$

If moreover d is real and the right-hand side is $\leq \varepsilon$, then $|\mathrm{Re}(P/Q) - d| \leq \varepsilon$.

LEAN: `ratio_enclosure, re_enclosure`

Proof sketch. The numerator identity is

$$P - dQ \;=\; \big(P - d^{-m}\big) - d\big(Q - d^{-(m+1)}\big),$$

since $d \cdot d^{-(m+1)} = d^{-m}$; so $|P - dQ| \leq \alpha + |d|\beta$. For the denominator, the reverse triangle inequality gives $|Q| \geq |d|^{-(m+1)} - \beta > 0$, which in particular makes $Q \neq 0$. Dividing, $|P/Q - d| = |P - dQ|/|Q|$ is at most the stated quotient. Taking real parts can only shrink the error: for d real, $|\mathrm{Re}(P/Q) - d| = |\mathrm{Re}(P/Q - d)| \leq |P/Q - d|$. □

The hypothesis $\beta < |d|^{-(m+1)}$ is exactly what keeps Q away from zero, and it is checkable. The two budgets α, β are the *only* route by which any approximation enters: the finite-bank truncation error and the jet tail both land there, and the theorem is indifferent to which contributed what. This is the ancestor of the argument in Gargantini and Henrici [7], where circular arithmetic on the same power-sum ratios yields inclusion discs for polynomial roots; the contribution here is that the two budgets are separated and the statement is machine-checked.

The counting routine enters the next statement as an interface, and the interface has exactly one requirement. It is worth isolating, because it is the whole of what the theorem asks of a count.

Definition 44 (counting routine)**.** A *counting routine* is a function $R : \mathbb{R} \to \mathbb{N}$ that is *additive along the axis between non-events*: whenever $x \leq y$ and $\Xi(x), \Xi(y) \neq 0$,

$$R(y) \;=\; R(x) + \mathrm{ord}(x, y).$$

LEAN: `CountRoutine`

Nothing else is required — not that R counts from zero, not that it agrees with $N(t)$, not that it is computed in any particular way. Any procedure whose increments are the true event counts qualifies, and the theorem below is stated relative to whichever one is supplied. This is what §19(iv) means when it says the certified index is the index *of the count that was given*.

Theorem 45 (the product theorem)**.** *Let R be a counting routine with $R(a)+1 = i$ and $R(b) = i$, let $\Xi(a), \Xi(b) \neq 0$ and $a \leq b$. Then Ξ has exactly one zero γ in (a, b); and if, for that γ, the data $y_0, P, Q, m, \alpha, \beta, \varepsilon$ satisfy the hypotheses of Theorem 43 with $d = y_0 - \gamma$, then*

$$\gamma \ \in \ \big[\, y_0 - \mathrm{Re}(P/Q) - \varepsilon, \ \ y_0 - \mathrm{Re}(P/Q) + \varepsilon \,\big].$$

LEAN: `gamma_enclosure, count_eq_one_of_endpoints, unique_event_of_count_eq_one, one_le_order_of_zero`

Proof sketch. Additivity (Definition 44) applied to $a \leq b$ gives $R(b) = R(a) + \mathrm{ord}(a, b)$, and the two supplied counts then force $\mathrm{ord}(a, b) = 1$. Now $\mathrm{ord}(a, b)$ is by definition the *sum* of the multiplicities of the events in (a, b), and every event has multiplicity at least 1; a sum of positive integers equal to 1 has exactly one summand, equal to 1. So there is exactly one zero γ of Ξ in (a, b), and it is simple — note that this comes from the count alone, with no derivative hypothesis and no appeal to Theorem 38. Applying Theorem 43 to that γ with $d = y_0 - \gamma$ bounds $|\mathrm{Re}(P/Q) - (y_0 - \gamma)|$ by ε, which rearranges to the stated interval. □

Two features of this statement carry the weight. The counting routine enters only as an interface: any procedure satisfying Definition 44 may be substituted, and the theorem does not ask how it is computed. And the event's multiplicity is *certified by the count*, not assumed — the passage from $R(b) - R(a) = 1$ to "one simple zero" is the sum-of-orders argument in the proof above, so no simplicity hypothesis is imported here or anywhere else in the paper.

It is worth being precise about what this buys, because it is easy to overstate. The enclosure of Theorem 42 obtains its count from the chain, whose hypotheses are sign evaluations and non-vanishing of Ξ'; the enclosure here obtains it from R, and inherits whatever R certifies. The two are independent routes to the same integer, and the paper proves both rather than choosing.

Measured conditioning

The enclosure is a theorem for any α, β; how *small* ε gets is a question about the jets, and it is governed by

$$\kappa \ = \ \frac{|y_0 - \gamma_1|}{|y_0 - \gamma_2|},$$

the distance ratio from the base point to the nearest and second-nearest ordinates. The error of A_m/A_{m+1} should contract by κ per order. Measured against the true ordinates at 50-digit precision (`extractor/checks/jet_sweep.py`), over ten base points with κ ranging from 0.0143 to 0.3426, the mean contraction over $m = 5, \ldots, 15$ agreed with κ to between 1.2% and 2.7% — confirmed at all ten, against a pre-registered threshold of 5%.

The law degrades exactly where it should. At a base point placed between two ordinates at nearly equal distance ($\kappa = 0.882$) the measured contraction was 1.29: the error does not decay at all, and it oscillates, because the two nearest contributions enter with opposite signs and there is no dominant term to extract. This is not a defect of Theorem 43, which remains true;

it is the statement that a badly placed base point yields a large ε. A large ε is a wide interval, never a wrong one.

Remark 46. This entire section is *off* the certification path. The enclosure of Theorem 42 — which ordinate, and that it is unique — is complete without any jet, any tail bound, or any conditioning hypothesis. What the jets buy is narrowness, and if they are badly conditioned the only consequence is that the interval is wide.

13 The extractor

13.1 The head law

The lane split is not free: it manufactures a stationary point, and that stationary point sets the cost. Writing the n-th term of the lane-balanced bank at height T, its phase is

$$\varphi(n) \;=\; \pi n - T\log(\Delta n),$$

the πn supplied by the alternating lane sign and the $T\log(\Delta n)$ by the phasor. Then $\varphi'(n) = \pi - T/n$, which vanishes at

$$n_* \;=\; T/\pi.$$

Below n_* the terms oscillate and cancel; at n_* they align. A head window that is tapered to zero at its edge therefore has to carry n_* strictly inside its flat core, or the truncation cuts the bank exactly where it is not cancelling. With a window whose flat core is the first half, this forces

$$X \;>\; 2T/\pi.$$

The head is linear in the ordinate, not $O(\sqrt{T})$. This is the single structural cost of the construction, and it is the price of the lane split: the Riemann–Siegel head is $O(\sqrt{T})$ precisely because its phase has no such stationary point in range.

This is a *design* rule: it says where to put the window, not what accuracy a given head achieves, and the two are worth separating. The script `extractor/checks/head_threshold.py` measures the smallest head $X_{\min}$ reaching a fixed relative accuracy, as a multiple of $2T/\pi$:

T	tol $= 10^{-6}$	10^{-9}	10^{-12}
500	1.12	1.37	1.69
1500	1.00	1.10	—
5000	0.95	0.99	—
15000	0.96	0.98	—

So the accuracy threshold tracks $2T/\pi$ to within a few percent at large T, but it is *not* approached from one side: at a loose tolerance and large height the rule is mildly conservative (ratio 0.95), and at a tight tolerance and small height it is not enough on its own (ratio 1.69). The dashes are not omissions. They record that a `float64` bank *cannot* reach 10^{-12} relative accuracy at those heights at any head length: the phase $T\log(\Delta n)$ carries absolute error of order $T\log(X)\cdot 2.2\times10^{-16}$, which floors the bank's relative accuracy near 10^{-13} at $T\sim10^3$ and near 3×10^{-10} at $T = 1.5\times10^4$. The `--tol` parameter therefore sets the head *length* via the law below; it does not by itself deliver the corresponding accuracy, and past the floor a longer head buys nothing. This is the same arithmetic limit that (O1) proposes to remove.

Past the threshold the convergence is fast and the right variable is the excess. Writing $s = \sqrt{\pi X/2 - T}$, the relative error of the truncated bank falls like $\exp(-cs)$. Measured at 60-digit working precision against $\Delta^{-s}(1-2^{1-s})\zeta(s)$ (`extractor/checks/head_law_decay.py`, `extractor/checks/head_law_moments.py`), over heights $T = 143, 500, 1500$, carrier scales $\Delta \in \{\pi/3, 1\}$ and moment orders $r = 0, \dots, 4$:

$$-\ln(\text{err})/s \;\in\; [1.400,\, 1.485] \quad \text{for } s \gtrsim 10,$$

across thirteen orders of magnitude of error. The dependence on the moment order is slight — the five orders $r = 0, \dots, 4$ span 0.085 at fixed s — and the dependence on Δ is nil to every digit printed. For s below about 10, that is for a head only just past the threshold, the effective rate is *larger* (up to 2.7 at $s \approx 5.6$), so the exponential law is approached from above and using a constant from the asymptotic band is the conservative choice. The implementation sets $X = 2(T+K)/\pi$ with $K = (\ln(1/\text{tol})/1.43)^2$; the constant 1.43 sits inside the measured band, and at $\text{tol} = 10^{-16}$ the difference between 1.43 and the band's lower end 1.40 is thirteen terms of head.

Remark 47 (why this had to be measured in high precision). A double-precision run cannot see this law at all. The bank summed in `float64` saturates near 10^{-13} relative, so its error stops falling once the head passes the threshold and the measured $-\ln(\text{err})/s$ then decays like $1/s$ — an artifact of the arithmetic, not a property of the bank. The figures above were therefore taken with the bank summed in `mpmath`; the reported thirteen orders of magnitude are what 60 digits of working precision will support, and no claim is made beyond them.

13.2 Algorithm

The extractor takes an index i and returns an interval.

Stage 1 (anchor). Invert the archimedean winding ϑ for a coarse ordinate: solve $\vartheta(T)/\pi + 1 = i$ by Newton on a smooth monotone function. Cost is polylogarithmic and the result is within a few mean spacings.

Stage 2 (bracket). Bisect on the predicate $\text{Count}(t) \geq i$, which is monotone in t. This is the one place the count enters, and it enters only as an *interface*: any routine satisfying Definition 44 may be substituted, and the certification in Stage 5 is stated relative to whatever it returns (§11, §12). Bisection rather than a fixed-step march is not a detail — see §13.3 on Lehmer's pair.

Stages 3–4 (moments and jets). Write v for the vector of site phasors $\left(x_n^{-s}\right)_n$ over the head, ℓ for the lane-balanced summation functional that forms the fiber from it (the alternating sum of Definition 16), and H for the log-weight operator of §1 (Theorem 60), so that $\ell(v) = F_\Delta(s)$ truncated to the head. One pass over the head evaluates the bank and, simultaneously, the log-weighted moments $M_r = \ell(H^r v)$, since the weights $\log x_n$ are already formed. A Bell (cumulant) recursion converts the moments into the log-jets $A_m = \sum_j (y_0 - \gamma_j)^{-m}$, and the ratio A_m/A_{m+1} locates the nearest event with error $O(\kappa^m)$, κ the ratio of distances to the nearest and next-nearest events. The whole jet vector costs one pass, not m.

Stage 5 (certificate). Narrow to a sign bracket around the jet estimate and return it. The returned interval is certified *given* the count, which is exactly the shape of the product

theorem, Theorem 45 — *not* of the capstone. The distinction matters and is set out in (O1): the implementation takes the count route, which carries no derivative hypothesis at all.

A noise floor is estimated from the head tolerance and the bank magnitude; a bracket narrower than the floor is reported as *not certified* rather than returned silently. That guard was added after an early version produced false certificates by narrowing past its own arithmetic.

13.3 Validation

The harness checks three things at every index: that the count interface actually reached $\mathrm{Count}(a) = i - 1$ and $\mathrm{Count}(b) = i$; that Stage 5 returned an interval rather than hitting its noise floor; and that the interval contains the reference ordinate. The reference is `mpmath`'s `zetazero`, which is an independent implementation — it is not derived from anything in this paper.

Remark 48 (two layers, and which one this table is). The word *certified* is used in two senses in this paper and they should not be run together. This section reports the `float64` extractor, which *locates*: it is fast, it is validated against an independent reference at every index below, and its internal guard is a measured noise floor, which is a heuristic. The certification proper is a separate layer, §20(O1): an Euler–Maclaurin evaluator with an explicit remainder, which proves the signs, the non-vanishing of Ξ' and the count on the intervals this layer produces. The table below is evidence that the locator works; it is not the certificate, and nothing in it is claimed to be.

sample	indices	failures
exhaustive $1 \le i \le 200$	200	0
random $201 \le i \le 2 \times 10^4$	90	0
spot checks to $i = 10^6$	5	0
Lehmer's pair $i = 6709, 6710$	2	0
total	297	0

Lehmer's pair [4] is the interesting one. The two ordinates are

$$\gamma_{6709} = 7005.062866174921, \qquad \gamma_{6710} = 7005.100564672656 :$$

a separation of 0.0377 against a local mean spacing of 0.8955, i.e. 0.042 of a mean spacing. A fixed-step bracket search with any step comparable to the mean spacing walks straight over both; bisection on the monotone count predicate does not, because it never needs to resolve the pair — it only needs the count to cross. Both are returned certified, and each contains its reference ordinate. This was a genuine failure of an earlier version (a fixed-step march at 0.30 spacings returned γ_{6711} for $i = 6710$), and it is the reason Stage 2 is stated as bisection on a predicate rather than as a search.

Carrier-scale invariance. The whole construction is supposed to be Δ-free, and this is testable rather than assumed: running $i = 1$ and $i = 100$ on $\Delta \in \{\pi/6, \pi/3, 1, 2, 30\}$ — a factor of $180/\pi \approx 57$ between the smallest and largest carrier scale — returns the same ordinates to every digit printed. That is Proposition 25 as a measurement.

14 Eigenvalues and the rank drop: reading the abscissa on the line

Everything so far has used the readout's *sign*, which is one bit per evaluation and is blind to the abscissa by construction. The moment vector of Stage 3 carries more than that, and this section extracts it. The claim is that the abscissa of a zero — whether it lies on the critical line at all — is recoverable from data computed *only on the line*, as the imaginary part of an eigenvalue.

14.1 The jets are a power sum over the fiber's zeros

Write $G(y) = F_\Delta(\frac{1}{2} + iy)$ for the fiber read in the ordinate variable: a meromorphic function of y whose zeros are those of ζ (non-trivial and trivial) and of the lane factor, and whose one pole — ζ's pole at $s = 1$ — sits at $y = -\frac{i}{2}$. Near a real base point y_0, for $m \geq 2$,

$$A_m \;=\; \frac{(-1)^{m-1}}{(m-1)!}\,\big(\log G\big)^{(m)}(y_0) \;=\; \sum_j \mu_j\, d_j^{\,m}, \qquad d_j = \frac{1}{y_0 - y_j}, \tag{1}$$

the sum over the zeros *and the pole* of G, with μ_j the multiplicity of a zero and $\mu = -1$ at the pole. The series converges absolutely for $m \geq 2$: the non-trivial zeros number $O(\log T)$ per unit ordinate, so $\sum_j |y_0 - y_j|^{-m} < \infty$, and the trivial zeros, the lane nodes and the pole sit at distance $\gtrsim y_0$ from a base point on the line. This is the m-th logarithmic derivative of a Hadamard product of genus one: for $m \geq 2$ the exponential factor and the convergence factors differentiate away and what remains is exactly the power sum. At order one it is *not* exact — an earlier version of this section said it was, and that was an erratum: $(\log G)'(y_0)$ carries, besides a symmetrically summed $\sum_j \mu_j/(y_0 - y_j)$, the constant $-i \log \Delta$ from the carrier scale and the constants of the Hadamard factor and of $\Gamma_{\mathbb{R}}$. That the carrier scale shows itself in the jets *only* at order one, and there as the shift $\log \Delta$, is Theorem 60(iii) read on the jets. For $m \geq 2$ the jets are an exact *power sum* in the reciprocal distances, and (1) is what makes the rest of this section possible: a power sum determines its nodes. The extractor uses $m \geq 2$ only (its default order is 9), and from a base point on the line the pole and the trivial zeros are nodes at distance $\gtrsim y_0$, contributing $O(y_0^{-m})$.

The A_m cost one pass. The bank evaluates $M_r = \ell(H^r v)$ for $r = 0, \ldots, 2K+1$ simultaneously (Stage 3), and a Bell recursion converts $M_r \mapsto A_m$; no additional evaluation of ζ, and no differentiation of anything, is involved.

14.2 The node dictionary

Which y_j occur is the point. G is $\Delta^{-s}(1 - 2^{1-s})\zeta(s)$ at $s = \frac{1}{2} + iy$, so its zeros are those of ζ together with those of the lane factor. Under $s = \frac{1}{2} + iy$ a zero of ζ at $\rho = \beta + i\gamma$ sits at

$$y \;=\; \gamma - i\big(\beta - \tfrac{1}{2}\big), \qquad \text{so} \qquad \operatorname{Im} y = 0 \iff \beta = \tfrac{1}{2}. \tag{2}$$

The imaginary part of the node is the signed distance of the zero from the critical line. A node on the real axis is an on-line zero; a node off it reports its own abscissa as $\beta = \frac{1}{2} - \operatorname{Im} y$. Nothing in the computation left the line to learn this.

14.3 The pencil

Recovering nodes from a power sum is classical: it is Prony's problem, taken here in the numerically stable matrix-pencil form of Hua and Sarkar [5]. Build the Hankel data matrix $Y = (A_{1+i+j})_{0\le i\le K,\,0\le j\le K}$, take its singular value decomposition, keep the K_r dominant right singular directions V, and form the shifted pair $(V_{1:}, V_{:1})$; the eigenvalues of $V_{1:}^{+}V_{:1}$ are the d_j, and $y_j = y_0 - 1/d_j$.

14.4 What the rank counts

The rank of Y is the number of *distinct* nodes. This is worth stating carefully because it is easy to misread, and because it is the instrument's one structural limitation. Measured directly on synthetic power sums with known nodes (`extractor/checks/rank_drop.py`):

configuration	numerical rank	σ_2/σ_1
$1, 2, 3, 4$ distinct nodes	$1, 2, 3, 4$	—
one node of multiplicity 2	1	1.3×10^{-18}
one node of multiplicity 1	1	1.3×10^{-18}

The last two rows are the limitation: a double node and a simple node give *bit-identical* singular-value profiles. Multiplicity enters (1) as the coefficient μ_j, not as an extra exponential, so the rank cannot see it. **The rank detects coincidence, not multiplicity.** A rank of 1 where the count says 2 means two events have merged to within the resolution; it does not certify a double zero, and we do not use it to.

The approach to a merger is quantitative. With two nodes at separation ε, the measured σ_2/σ_1 was $4.58 \times 10^{-3},\ 4.24 \times 10^{-5},\ 4.20 \times 10^{-7},\ 4.20 \times 10^{-9}$ at $\varepsilon = 10^{-1}, \ldots, 10^{-4}$: a clean

$$\sigma_2/\sigma_1 \ \approx\ 4.2\,\varepsilon^2,$$

so in double precision, where the floor is near 10^{-16}, two events remain distinct down to $\varepsilon \approx 5 \times 10^{-9}$ and are indistinguishable below it. That number is the resolution of the instrument, and it is a property of the arithmetic rather than of the geometry.

14.5 The calibration standard the fiber carries

An instrument that claims to measure the abscissa must be checked against zeros whose abscissa is known and *is not* $\frac{1}{2}$. The fiber supplies them. The lane factor $1 - 2^{1-s}$ vanishes exactly when $1 - s \in \frac{2\pi i}{\log 2}\mathbb{Z}$, that is on the line $\operatorname{Re} s = 1$, and under (2) these sit at

$$y \ =\ -\frac{2\pi k}{\log 2} - \frac{i}{2}, \qquad k \in \mathbb{Z}: \qquad \operatorname{Im} y = -\tfrac{1}{2} \text{ exactly}, \quad \text{spacing } \frac{2\pi}{\log 2} = 9.064720. \tag{3}$$

These are genuine off-line nodes, in closed form, carried by the object itself — not planted, not synthetic, and not adjustable. A working pencil must place them at $\operatorname{Im} y = -\frac{1}{2}$, equivalently report $\beta = 1$ for them.

Measured (`extractor/checks/pencil_lane_control.py`), at an anchor $y_0 = 17.5$ placed between the first two ordinates so that no node dominates, with $K = 8$:

returned node	identification	error
$18.129441 - 0.500000\,i$	lane, $k = -2$ of (3)	2.0×10^{-14}
$14.134707 + 0.000026\,i$	γ_1, on-line	1.8×10^{-5}
$21.022108 + 0.000122\,i$	γ_2, on-line	6.8×10^{-5}

From one moment vector computed on the critical line, the pencil returns $\beta = 1$ for the lane zeros to fourteen digits and $\beta = \frac{1}{2}$ for ζ's zeros, and it separates the two populations. At $y_0 = 12$ the $k = -1$ lane node comes back as $9.064720 - 0.500005\,i$, against the predicted $9.064720 - \frac{i}{2}$.

This control is what makes the abscissa reading falsifiable rather than decorative, and it has already done work: it refuted two successive implementations of the pencil — a wrong data-matrix shape, and a spurious conjugation that returned $\overline{y}$ in place of y — both of which had produced clean-looking certificates while inverting precisely the off-line information the pencil exists to read.

14.6 Lehmer's pair, resolved

The hardest close pair in the low range is $\gamma_{6709}, \gamma_{6710}$, separated by 0.0377, or 0.042 of the local mean spacing. At an anchor between them (`extractor/checks/lehmer_rank.py`), with $K = 4$, the pencil returns rank 2 with $\sigma_2/\sigma_1 = 1.6 \times 10^{-10}$ — far above the floor — and the two nodes

$$7005.100564673 + 2.1 \times 10^{-12} i, \qquad 7005.062866175 - 2.7 \times 10^{-12} i,$$

matching the reference ordinates to 9.1×10^{-13} and 3.2×10^{-11}, with both abscissae $\beta = \frac{1}{2}$ to 3×10^{-12}. So the pair is resolved as two distinct events and both are certified on-line by the instrument, at a separation forty times finer than would trouble it.

14.7 The singular values, certified

The rank reading above was a measurement. It need not be, and making it a certificate is worth doing because the rank is the one part of this instrument that carries a *count*, and counts are what the rest of the paper certifies.

The chain is entirely within the machinery of §20, with one new element. Certified values of Ξ give certified *Taylor coefficients* at an anchor c by the trapezoid rule on a circle $|z - c| = R$: the n-point rule returns the aliased coefficient $\sum_{i \geq 0} a_{k+in} R^{k+in}$, and bounding $|\Xi| \leq M'$ on a larger disc of radius $R' > R$ gives $|a_j| \leq M'/R'^{\,j}$ by Cauchy, hence

$$\left| a_k R^k - \widehat{a}_k \right| \;\leq\; \frac{M' \, (R/R')^{k+n}}{1 - (R/R')^n},$$

an explicit geometric bound. The point of routing through Ξ rather than Ξ'/Ξ is that Ξ is *entire*: no lower bound on $|\Xi|$ is needed anywhere, and the region bound (O5) supplies M'. Complex interval arithmetic then carries the coefficients through $f_k = \Xi^{(k)}/\Xi$ and the cumulant recursion to certified jets A_m; Weyl's inequality, $|\sigma_i(H) - \sigma_i(H_{\text{true}})| \leq \|E\|_2 \leq \|E\|_F$, turns those into certified singular values.

Measured (`extractor/checks/certified_rank.py`), at three anchors with $K = 3$: the error norm $\|E\|_F$ came out between 3.7×10^{-29} and 5.3×10^{-27}, every singular value is enclosed to that width, and *every consecutive pair of enclosures separates* — so the ordering of the spectrum is proved, not observed. For the anchor $c = 17.5$: $\sigma_1 = 6.773 \times 10^{-1}$, $\sigma_2 = 8.088 \times 10^{-2}$, $\sigma_3 = 1.970 \times 10^{-3}$, $\sigma_4 = 1.992 \times 10^{-5}$, each with radius 2.6×10^{-28}.

14.8 Propose and verify: the node locations, certified

Theorem 50 certifies the rank for the contour-restricted moments. It says nothing about *where* the nodes are, and that is what the pencil supplies: the node locations, abscissa included. Those need not be trusted either — they can be *verified*:

(1) the contour certifies the count K in the band, with no pencil involved;

(2) the pencil, run on certified jets, *proposes* K nodes with abscissae;

(3) the certified machinery of §20(O1) verifies each proposal — a certified sign change with $\Xi' \neq 0$ on a small interval about it;

(4) if the number verified equals K, every node is accounted for.

The tail never enters: step (1) counts and step (3) locates, and neither consults the pencil's own error. The script `checks/pencil_verify.py` measures this at the anchors the extractor actually uses, with $K = 2$ and a $\pm 10^{-4}$ verification window:

i	proposed node	$\operatorname{Im} y$	node error	verified
1	14.1347251374	2.8×10^{-36}	4.3×10^{-9}	yes
10	49.7738359156	-2.1×10^{-29}	3.4×10^{-6}	yes
100	236.524229631	4.1×10^{-26}	3.4×10^{-8}	yes
1000	1419.42248108	-6.5×10^{-24}	1.3×10^{-7}	yes

All four verify, and $\operatorname{Im} y$ — the abscissa offset — comes back at 10^{-24} or smaller in every case.

Remark 49 (the conditioning is not incidental). A first attempt ran this at $c = 17.5$, midway between γ_1 and γ_2, and *all* proposals failed verification: the pencil returned 14.068, 20.872, 27.666 against true ordinates 14.135, 21.022, 25.011. That is the $\kappa \approx 0.88$ case of §12 — no nearest node dominates — and the failure is the conditioning law doing exactly what it says. The extractor never uses such an anchor: Stage 2 bisects first, which is what makes Stage 4 well conditioned. The propose-and-verify pipeline inherits that dependence, and a verification window must be scaled to the conditioning, not fixed.

14.9 The rank, certified

An earlier version of this section left the *rank* uncertified. The obstruction was that $A_m = \sum_j d_j^m$ runs over *all* zeros, so the exact Hankel matrix is not of finite rank and "rank K" is only a statement about a threshold; certifying it looked to require an explicit Riemann–von Mangoldt constant to bound the tail $\sum_{|c-\gamma_j|>D} d_j^m$.

No such constant is needed. The tail is an artifact of summing over all zeros, and restricting to a contour removes it by construction.

Theorem 50 (the rank)**.** *Let C be a contour carrying no zero of Ξ, enclosing exactly K zeros of Ξ, all simple and at distinct points. Put*

$$s_m \;=\; \frac{1}{2\pi i}\oint_C z^m \,\frac{\Xi'(z)}{\Xi(z)}\,dz, \qquad m = 0, 1, 2, \ldots$$

Then $s_m = \sum_{j=1}^{K} \gamma_j^m$, and the Hankel matrix $H = (s_{i+j})_{0 \le i,j \le K}$ has rank exactly K.

Proof. z^m is entire, so by the residue theorem the only contributions are the zeros inside C; each being simple, $s_m = \sum_j \gamma_j^m$. Let V be the $(K+1) \times K$ matrix $V_{i,k} = \gamma_k^i$. Then $s_{i+j} = \sum_k \gamma_k^i \gamma_k^j$, that is $H = VV^{\mathsf{T}}$. V is a Vandermonde in K *distinct* nodes with $K+1$ rows, hence of full column rank K. Over $\mathbb{C}$ that is not yet enough on its own — VV^{T} is not VV^*, and AA^{T} can have smaller rank than A — but V^{T} has full row rank K, so it maps $\mathbb{C}^{K+1}$ *onto* $\mathbb{C}^K$, on which V is injective; hence $\operatorname{rank} H = \operatorname{rank}(VV^{\mathsf{T}}) = K$. □

Each hypothesis is discharged by machinery already built: no zero on C by a certified positive lower bound for $|\Xi|$ there — sample, then fill the gaps with $|\Xi(z)| \geq |\Xi(z_k)| - \frac{L}{2} M/\rho$, the same device as the sound winding criterion; exactly K zeros inside by the certified argument principle; simple and distinct by §16.

Measured (`extractor/checks/rank_certified.py`):

centre	R	K	s_0	s_1	Hankel rank
14.1347	0.30	1	1.0	14.1347251417	1
17.6	3.60	2	2.0	35.1567647805	2
19.5	6.00	3	3.0	60.1676223607	3

s_0 returns the count as an integer, s_1 matches the sum of the reference ordinates to twelve digits, and the trailing singular value falls to between 10^{-18} and 10^{-20} — the arithmetic floor — in every case. On the first contour, which passes within 0.3 of γ_1, the lower bound for $|\Xi|$ needed $n = 8192$ samples to come out positive (3.6×10^{-4}); a contour near a zero costs sampling, which is the only price this route pays.

Remark 51 (what changed, and what did not). The full-jet Hankel is still not of finite rank, and its numerical rank is still a threshold statement — that was never wrong. What was wrong was the inference that certifying *a* rank therefore required a zero-density input. The rank question is well posed for the contour-restricted moments and ill posed for the unrestricted ones, and posing it correctly removes the need for the constant rather than supplying it.

14.10 The inertia: what the sign of the Hankel spectrum counts

Theorem 50 restricts the moments to a contour, $s_m = \sum_j y_j^m$ over the K nodes enclosed, and reads the rank. The same matrix carries a second integer.

Lemma 52 (inertia)**.** *Let the enclosed nodes consist of r real ones $y_1, \ldots, y_r$ and q conjugate pairs $z_j, \bar{z}_j$, all distinct, and let $H_n = (s_{i+j})_{0 \leq i,j < n}$ with $n \geq r + 2q$. Then the inertia of H_n is $(r+q,\ q,\ n-r-2q)$: the number of negative eigenvalues is exactly the number of off-line pairs enclosed, and $H_n \succeq 0$ if and only if $q = 0$.*

Proof. With $v(y) = (1, y, \ldots, y^{n-1})^{\mathsf{T}}$, $H_n = \sum_i v(y_i) v(y_i)^{\mathsf{T}} + \sum_j \big[v(z_j) v(z_j)^{\mathsf{T}} + v(\bar{z}_j) v(\bar{z}_j)^{\mathsf{T}} \big]$. Writing $v(z_j) = A_j + iB_j$ with A_j, B_j real, the bracket is $2A_jA_j^{\mathsf{T}} - 2B_jB_j^{\mathsf{T}}$ — the minus sign is the whole mechanism — so $H_n = UJU^{\mathsf{T}}$ with $U = [\, v(y_1) \cdots v(y_r)\ \ A_1\, B_1 \cdots A_q\, B_q \,]$ and $J = \operatorname{diag}(1, \ldots, 1,\ 2, -2, \ldots, 2, -2)$. U has full column rank $r + 2q$ (a Vandermonde in distinct nodes followed by an invertible change of columns), and Sylvester's law of inertia gives the count. □

So the contour moments — computed from Ξ'/Ξ alone, with no zero located — carry the number of zeros off the line inside the contour as the negative inertia of a real symmetric matrix.

Measured (`extractor/checks/operator_positivity.py`) with the certified moments of §14.9 and eigenvalue enclosures by Weyl's inequality:

centre	R	K	$\max\lvert\mathrm{Im}\, s_m\rvert$	$\lambda_{\min}/\lambda_{\max}$
14.135	0.30	1	3.3×10^{-31}	$+7.1\times10^{-39}$
17.600	3.60	2	2.1×10^{-22}	$+6.2\times10^{-38}$
19.500	6.00	3	5.0×10^{-19}	-5.8×10^{-40}

As a positive control (`extractor/checks/positivity_control.py`), a planted conjugate pair at distance $\beta-\frac{1}{2}$ from the line produces a negative eigenvalue whose size scales as $(\beta-\frac{1}{2})^2$ — 5.4×10^{-10} at 0.5 down to 2.1×10^{-21} at 10^{-6} — against a rounding floor of 8×10^{-59}.

Remark 53 (status). This is a *reading*, not yet an independent certificate. An eigenvalue enclosure $\lambda_{\min}\geq-\varepsilon$ does not decide $q=0$ by itself: a pair close enough to the line contributes a negative eigenvalue smaller than ε. To make the inertia a certificate one needs a *separation* — a proved lower bound on that negative eigenvalue in terms of the distance from the line and the conditioning of U — and no such bound is proved here. On the three windows above, $q=0$ is established by Theorem 56 with Corollary 59; the inertia reading agrees with it to forty digits, which is a consistency check of Lemma 52 and of the certified moments, and is reported as that.

14.11 Status

This section is *off* the certification path of §11, and nothing above depends on it. It is a finite-rank fit to a truncated moment vector in double precision; the errors reported are measured, not bounded a priori, and obligation (O3) applies to them. What is established is that the abscissa is *present* in the on-line moment data, that a standard eigenvalue method extracts it, and that the extraction is calibrated against an off-line set the fiber carries in closed form. Two routes would make it a certificate rather than a measurement: interval arithmetic on the Hankel matrix, which would turn the singular-value gap into a certified rank; and an a priori bound on the finite-bank error in (1), which is the same α,β budget as (O3). Neither is done here.

15 The model for a general Dirichlet L-function

The carrier does not change and the warp does not change. Exactly two things do: the lane structure and the weld gauge.

The lanes are the conductor's channels. For a primitive character χ the fiber is $\sum_n \chi(n)x_n^{-s}$, the site weights supplied by the character. The alternating split used throughout §6 is the special case of χ trivial, reached through $\eta(s)=(1-2^{1-s})\zeta(s)$: the two lanes are then the even and odd sites, and the $\pm$ is the value of the non-principal character mod 2. So the lane split is not a device introduced for ζ; it is what the conductor supplies, seen at conductor 2.

The archimedean factor carries the parity. The gamma factor is $\Gamma_{\mathbb{R}}(s+a)$ with $a\in\{0,1\}$ the parity of χ, and the weld gauge is read off from it exactly as before.

The root number is absorbed, not assumed away. The completed function satisfies $\Lambda(s,\chi)=\varepsilon(\chi)\Lambda(1-s,\bar\chi)$, so on the critical line $\Lambda(\frac{1}{2}+it,\chi)=\varepsilon(\chi)\overline{\Lambda(\frac{1}{2}+it,\chi)}$ at every t. A value fixed by such an anti-involution lies on a fixed ray, at half the argument of ε. Hence the readout is real for *every* primitive character, not only the self-dual ones, provided the gauge is rotated by $\arg\varepsilon(\chi)/2$ — and that rotation is a constant, computed once from the Gauss sum, not a function of t.

Proposition 54 (the weld absorbs the root number)**.** *Let $\phi \in \mathbb{R}$. A value satisfying the weld anti-involution $z = e^{i\phi}\bar{z}$ lies on the $\phi/2$ ray: $z\,e^{-i\phi/2} \in \mathbb{R}$. With $\phi = \arg\varepsilon(\chi)$ this is the statement that $\Lambda(\frac{1}{2}+it,\chi)$, rotated by half the root number, is real at every t.*

LEAN: `HingeKernel.weld_pins_half_phase, UnitMidpoint.conj_fixed_iff`

Proof sketch. Put $u = e^{i\phi/2}$, so $e^{i\phi} = u^2$ and $\bar{u} = u^{-1}$. The relation $z = u^2\bar{z}$ reads $zu^{-1} = \overline{zu^{-1}}$, i.e. $ze^{-i\phi/2}$ is fixed by conjugation, i.e. real. For the application, the functional equation $\Lambda(s,\chi) = \varepsilon(\chi)\Lambda(1-s,\bar{\chi})$ combined with $\Lambda(s,\bar{\chi}) = \overline{\Lambda(\bar{s},\chi)}$ gives, at $s = \frac{1}{2}+it$, $\Lambda(\frac{1}{2}+it,\chi) = \varepsilon(\chi)\overline{\Lambda(\frac{1}{2}+it,\chi)}$, which is the hypothesis with $\phi = \arg\varepsilon(\chi)$. □

The ray is determined only modulo π — both $\pm e^{i\phi/2}$ solve $u^2 = e^{i\phi}$ — which is a choice of overall sign for the readout and affects nothing below, since every statement is about zeros and sign *changes*. Computed directly from the Gauss sums (`extractor/checks/root_number_rays.py`): the order-4 character mod 5 gives the ray 0.088104π and the order-6 character mod 7 gives 0.186840π, with $|\varepsilon| = 1$ to machine precision in both cases and the conjugate characters mirrored about 0.

Remark 55. Everything downstream — the multiplicity sign law, the off-line pairing, the interval count, the count of one from a non-vanishing derivative, the chain — is stated for a real readout and therefore transfers verbatim once the readout is made real. The off-line pairing in particular survives non-self-duality: composing the functional equation with conjugation returns χ to itself, since $\Lambda(s,\chi) = \varepsilon(\chi)\overline{\Lambda(1-\bar{s},\chi)}$, so $\Lambda(\cdot,\chi)$ vanishes at s exactly when it vanishes at $1-\bar{s}$ — the reflection in the critical line, which is again $z \mapsto \bar{z}$ in the ordinate variable. The pairing is between zeros of the *same* function, not between χ and $\bar{\chi}$.

The formalisation in §21 is carried out for ζ; the general case changes the lane weights and the ray, and nothing else. Proposition 54 is itself machine-checked, in two further modules (`HingeKernel.lean`, `UnitMidpoint.lean`); the transfer of the downstream chain to a general primitive χ is not.

16 A certified on-line verification, carried out in the ordinate variable

Everything in §§7–11 runs on the readout's *sign*, and §19(iv) records the price: a sign bit cannot tell an on-line zero from a conjugate off-line pair. That limitation is real, and it is a limitation of the chart. This section removes it, and the removal is geometric.

The radial reading. Under $s = \frac{1}{2}+iz$ a zero $\beta+i\gamma$ sits at $y = \gamma - i(\beta-\frac{1}{2})$, so

$$|\operatorname{Im} z| \le h \qquad \Longleftrightarrow \qquad \left|\beta - \tfrac{1}{2}\right| \le h.$$

Off the critical line is off the real axis. The distance from the line is a *radial* coordinate in the ordinate variable, and a contour integral over the band $|\operatorname{Im} z| \le h$ counts exactly the zeros within h of the line — a quantity the one-dimensional readout has no way to express, because the descent of §5 discards precisely this direction.

Theorem 56 (band verification)**.** *Let $a < b$ and $h > 0$, let Ξ have no zero on the boundary of $[a,b]\times[-h,h]$, and suppose*

(i) the argument principle over $[a,b]\times[-h,h]$ certifies N zeros of Ξ in the band, counted with multiplicity, and

(ii) the readout has N certified sign changes on $[a,b]$, each isolated to its own subinterval, the N subintervals pairwise disjoint, on each of which $\Xi' \neq 0$ is certified.

Then every zero of ζ with ordinate in (a,b) and $|\beta-\frac{1}{2}|\le h$ lies on *the critical line and is simple.*

Proof. By (ii) and Theorem 38, each of the N subintervals contains exactly one event, and it is *simple*; these are N distinct real zeros of Ξ lying in (a,b), hence in the band, and being simple they contribute exactly N to the count of (i). By (i) the band contains exactly N zeros with multiplicity. So the band contains no further zero at all, and in particular none with $0<|\operatorname{Im} z|\le h$.

The boundary hypothesis is not an extra burden in practice: the sound step criterion of §20(O5)(a) requires $|\Xi(z_k)|$ to exceed a positive bound at every sample, so a zero on the contour makes the count fail to certify rather than certify wrongly. □

Both hypotheses are discharged by the machinery of §20: (i) by the sound winding criterion, which needs the explicit region bound (O5); (ii) by the certified signs and the certified derivative. Measured (`extractor/checks/band_verification.py`), at $h=\frac{1}{4}$ — that is, $\beta\in[\frac{1}{4},\frac{3}{4}]$:

window	band count	simple real zeros	evaluations	verdict
$[10,16]$	1	1/1	17 108	certified
$[10,23]$	2	2/2	32 042	certified
$[10,26]$	3	3/3	42 290	certified

Remark 57 (what this is, and what it is not). It is a *window* statement with a *band*: it certifies that no zero in that window has left the line by as much as h, and it says nothing outside the window or beyond the band. It is also not new as a capability — this is the conclusion Turing's method reaches, in Platt's rigorous form [12], far faster and far higher; the evaluation counts above are frank about the cost.

What it is, is the first statement in this paper that the chart cannot make. The sign-based chain is blind to the abscissa by construction; the band count sees it because the ordinate variable has a direction that the descent throws away, and the carrier supplies the object whose contour is being integrated. The geometry is not decorating an analytic argument here — it is supplying the coordinate in which the question can be posed at all.

The band, removed. The band in Theorem 56 is a limitation of the instrument, not of the statement: the region bound of §20(O5) is a Stirling bound and needs $\operatorname{Re}(s/2)>0$, so a certified contour in the ordinate variable cannot reach $|\operatorname{Im} z|=\frac{1}{2}$. What the contour cannot see is the sliver $\beta\in(0,\frac{1}{2}-h)\cup(\frac{1}{2}+h,1)$, and by $\xi(s)=\xi(1-s)$ the left piece is the reflection of the right. That region is not near the critical line; it is near $\operatorname{Re} s=1$, where ζ is bounded away from zero on any bounded range of ordinates — a finite computation in the s-plane, with no Γ-factor and no Stirling bound in it.

Proposition 58 (the sliver)**.** *Let $0<Y<\frac{1}{2}$, $a<b$ and $S_1>1$. Suppose*

(a) on the box $\sigma\in[\frac{1}{2}+Y,S_1]$, $t\in[a,b]$, a grid of certified enclosures of ζ, together with a certified $L\ge\sup|\zeta'|$ over the box, gives $|\zeta|-\mathrm{rad}>L\rho$ at every node, ρ the covering radius of a grid cell; and

(b) $|\zeta(s)| \geq \zeta(2\sigma)/\zeta(\sigma) > 0$ *for* $\sigma \geq S_1$*, which is the Euler product.*

Then ζ *has no zero with ordinate in* $[a, b]$ *and* $|\beta - \frac{1}{2}| > Y$*.*

Proof. (a) and (b) exclude zeros with $\sigma \geq \frac{1}{2} + Y$ in the ordinate range: every point of the box is within ρ of a node, so $|\zeta| \geq (|\zeta(\text{node})| - \text{rad}) - L\rho > 0$ there. A zero with $\beta < \frac{1}{2} - Y$ would give, by the functional equation and conjugation, a zero at $1 - \beta > \frac{1}{2} + Y$ with the same ordinate, which is excluded. The constant L is Cauchy's $\sup|\zeta'| \leq M/\delta$, with M an Euler–Maclaurin bound for $|\zeta|$ on the box enlarged by δ, every term bounded in modulus; the node enclosures are those of §20(O1). □

Corollary 59 (band-free window)**.** *If Theorem 56 certifies the window* $[a, b]$ *at band* h *and Proposition 58 certifies it at* $Y = h$*, then every zero of* ζ *with ordinate in* (a, b) *lies on the critical line and is simple.*

Measured (`extractor/checks/strip_sliver.py`); the decision rule was fixed before the run — a single node failing $|\zeta| - \text{rad} > L\rho$ reports a refutation and does not refine:

Y	ordinates	grid	L	$L\rho$	worst $\lvert\zeta\rvert$	ratio
0.45	$[13.4, 25.6]$	11×1232	25.23	0.1696	0.3016	1.78
0.40	$[13.4, 25.6]$	17×1361	27.87	0.1753	0.2735	1.56
0.45	$[9, 31]$	11×2352	26.72	0.1743	0.3015	1.73
0.25	$[10, 26]$	85×4525	39.59	0.0989	0.1806	1.83

The last row is at the band of the table above ($h = \frac{1}{4}$) and covers its three windows, so Corollary 59 applies to each of them: *every zero of* ζ *with ordinate in* $(10, 26)$ *lies on the critical line and is simple.* That is, again, a conclusion Turing's method reaches — and the companion Lean library already carries it to ordinate 371.5 (`online_below_743o2`) — and it costs some 385 000 certified evaluations of ζ on the sliver alone, the worst node sitting beside the first zero at $\sigma = 0.752$, $t = 14.13$. What is new is only that the geometric certificate now says it without a qualifier.

17 The Hamiltonian of the fiber, and the three-dimensional zeta

§1 named the warp's radial generator — the log-weight operator H — and asserted $H_\Delta = H + \log\Delta \cdot I$. A generator is an infinite-dimensional object, and the sentence is not a statement until a Hilbert space, a domain and a self-adjointness theorem are supplied; Stone's theorem needs exactly that. This section supplies them, and then reads off what the operator does: the fiber is its flow, the one-dimensional L-function is its trace, and the chart's zero is an event in the flow. Every statement is unconditional and machine-checked; nothing here bears on where the zeros are, and Remark 64 says so at the end.

17.1 The log-weight operator

Write $\mathcal{H} = \ell^2(\mathbb{N}, \mathbb{C})$, the index n standing for the site x_{n+1} (there is no site at the origin), and e_n for the n-th site phasor, the sequence with a 1 at n and 0 elsewhere. For a real weight w let M_w be multiplication by w on its *maximal* domain $\{v \in \mathcal{H} : w \cdot v \in \mathcal{H}\}$. The log-weight operator and its carrier-scaled form are

$$H = M_{\log(n+1)}, \qquad H_\Delta = M_{\log(\Delta(n+1))}.$$

Theorem 60 (the log-weight operator)**.** *(i) M_w is self-adjoint on its maximal domain — $M_w^\dagger = M_w$, domain and values. In particular H and every H_Δ are self-adjoint.*

(ii) The site phasors are orthonormal with dense span, a Hilbert basis of $\mathcal{H}$, and each is an eigenvector: $He_n = \log(n+1)\, e_n$.

(iii) For $\Delta > 0$, H_Δ and H have the same domain and $H_\Delta v = Hv + \log\Delta \cdot v$ on it: the carrier scale shifts the generator by a constant and does nothing else.

(iv) H is unbounded: no C satisfies $\|Hv\| \le C\|v\|$ on the domain.

LEAN: `CarrierReadout.weightMul_isFormalAdjoint, CarrierReadout.weightDomain_dense, CarrierReadout.adjointDomain_le, CarrierReadout.weightMul_isSelfAdjoint, CarrierReadout.logWeight_isSelfAdjoint, CarrierReadout.carrierLogWeight_isSelfAdjoint, CarrierReadout.phasor_orthonormal, CarrierReadout.phasor_total, CarrierReadout.phasorBasis, CarrierReadout.H_phasor, CarrierReadout.carrierLogWeight_eq_shift, CarrierReadout.logWeight_not_bounded`

Proof sketch. (i) has the two halves the statement needs: symmetry on a dense domain — the phasors lie in every maximal domain and span a dense subspace — and maximality, that a vector in the domain of the adjoint is already in the maximal domain, read off against each phasor. The second half is what fails on a smaller domain: the restriction of H to finitely supported vectors is symmetric and not self-adjoint. (ii) is the ℓ^2 basis. (iii) is $\log(\Delta(n+1)) = \log\Delta + \log(n+1)$ together with the observation that adding a constant to a weight does not change the maximal domain; the domain equality is `CarrierReadout.HΔ_domain` (the name carries a Greek delta). (iv): $\|He_n\| = \log(n+1)$ is unbounded. □

The domain is not a formality. Self-adjointness is a property of an operator *on a domain*, and where §1 calls H a generator the maximal domain is the one that must be meant. And the eigenvalues are the geometry's, twice over: by §4, $\log x_n$ is the ordinate at which the climber reaches the site, so H is the climber read on the sites; and by Theorem 4 it is, up to the affine freedom, the only continuous weight for which the carrier scale is a gauge.

17.2 The warp is the functional calculus of H

Let e^{-sH} denote multiplication by $(n+1)^{-s}$ on its maximal domain.

Theorem 61 (the generator)**.** *The eigenvalue of e^{-sH} on e_n is $\exp(-s\lambda_n)$ with $\lambda_n = \log(n+1)$ the eigenvalue of H: on the total eigenbasis of Theorem 60, e^{-sH} is the functional calculus of H. It satisfies $e^{-sH}e^{-s'H} = e^{-(s+s')H}$ on the eigenbasis, and for $s = -it$ each phasor is rotated by the unimodular $\exp(it\log(n+1))$: the warp on the critical line is the unitary group of H.*

LEAN: `CarrierReadout.heat_phasor, CarrierReadout.heat_eigenvalue_eq_exp, CarrierReadout.heat_eigenvalue_add, CarrierReadout.norm_line_phase`

This is the precise content of "the warp is scaling by a function of unit modulus whose radial generator is H": the function is $x_n \mapsto x_n^{-s} = e^{-s\log x_n}$, the generator is H, and unit modulus is the case $\operatorname{Re} s = 0$.

17.3 The one-dimensional function is the trace

Theorem 62 (ζ is the partition function of H)**.** *For* $\operatorname{Re} s > 1$,

$$\sum_{n\geq 0} \langle e_n,\ e^{-sH} e_n \rangle \ =\ \zeta(s), \qquad \sum_{n\geq 0} \Lambda(n+1)\, \langle e_n,\ e^{-sH} e_n \rangle \ =\ -\frac{\zeta'}{\zeta}(s).$$

The eigenvalue of H at the site $x = n+1$ decomposes over the primes, $\log(n+1) = \sum_{p|n+1} v_p(n+1)\log p$*, so* $H = \sum_p \log p \cdot N_p$ *with N_p the p-adic valuation at each site and each* $\log p \cdot N_p$ *self-adjoint; and the trace factorises accordingly,* $\operatorname{Tr} e^{-sH} = \prod_p (1-p^{-s})^{-1}$*, each factor the geometric partition function* $\sum_k (p^{-s})^k$ *of the single clock* $\log p \cdot N_p$*. Finally the entire readout is the completed trace: for* $\operatorname{Re} s > 1$*,*

$$\xi(s) \ =\ \frac{s(s-1)}{2}\, \Gamma_{\mathbb{R}}(s)\ \operatorname{Tr} e^{-sH},$$

so $\Xi(z) = -2\,\xi(\frac{1}{2} + iz)$ *(Definition 26) is -2 times the completed partition function of H, read in the ordinate variable.*

LEAN: `CarrierReadout.inner_phasor_heat`, `CarrierReadout.zeta_eq_heatTrace`, `CarrierReadout.vonMangoldt_heatTrace_eq`, `CarrierReadout.logWeight_eq_sum_primeClocks`, `CarrierReadout.primeClock_isSelfAdjoint`, `CarrierReadout.primeClock_phasor`, `CarrierReadout.heatTrace_eulerProduct`, `CarrierReadout.primeClock_partition`, `CarrierReadout.riemannXi_eq_completed_heatTrace`

The first identity is the one Bost and Connes read off their system [13], here proved of this paper's own operator; the second says that the prime side of the explicit formula is the trace of a function of H; the third is the Euler product carried by the operator rather than by the series — the multiplicative structure of the sites is in H, as a sum of commuting self-adjoint prime clocks. These are statements about the object: what the chart reads off the generator's flow. Their analytic halves — the Dirichlet series, the von Mangoldt series, the Euler product — are Mathlib's theorems (§21), and the proofs consist of identifying each as a trace over the phasor basis. The next subsection is the object those traces are taken of.

17.4 The three-dimensional zeta is the bank, and the chart's zero is an event

The scalar identities above are shadows of a vector one. The *bank* at height s is the vector of site phasors,

$$B(s) \ =\ e^{-sH}\mathbb{1} \ =\ \left(x_n^{-s}\right)_n,$$

$\mathbb{1}$ the all-ones state — every phasor at unit magnitude at height zero, $B(0) = \mathbb{1}$. This is the three-dimensional object of §4; the one-dimensional L-function is a *chart* of it, a linear functional applied to the bank.

Theorem 63 (the bank)**.** *(i)* H generates the bank. $B(s+s') = e^{-s'H} B(s)$*, componentwise for all s, s', and as vectors of $\mathcal{H}$ whenever* $\operatorname{Re} s > \frac{1}{2}$ *and* $\operatorname{Re} s' \geq 0$*.*

(ii) On the critical line the bank is a unitary orbit of one state. $B(\frac{1}{2} + it) = U(t)\, B(\frac{1}{2})$ *with $U(t) = e^{-itH}$ rotating each phasor by a unimodular phase.*

(iii) The critical line is the ℓ^2-boundary of the bank. $B(s) \in \ell^2$ *if and only if* $\operatorname{Re} s > \frac{1}{2}$*, and* $\|B(s)\|^2 = \sum_n x_n^{-2\operatorname{Re} s} = \zeta(2\operatorname{Re} s)$*, which diverges at the line through the pole at* 1*.*

(iv) Every L-series is a chart of the same bank. *For an L-series $L(s,f)=\sum_n f(n)n^{-s}$ convergent at s, $L(s,f)=\sum_n f(x_n)\,B(s)_n$; in particular the plain chart $\sum_n$ reads $\zeta(s)$ off the bank for* $\operatorname{Re} s>1$.

(v) The lane-balanced chart reads the fiber. *The chart of Definition 16 — the phasors summed in pairs, in the order they enter — reads $\eta(s)=(1-2^{1-s})\zeta(s)$ off the bank for every s with* $\operatorname{Re} s>0$*; this is the fiber F_Δ at carrier scale $\Delta=1$, the scale entering only as the factor Δ^{-s} of Theorem 60(iii).*

(vi) The chart's zero is an event. *In the strip* $0<\operatorname{Re} s<1$, $\zeta(s)=0$ *if and only if $B(s)$ lies in the kernel of the lane-balanced chart.*

LEAN: `CarrierReadout.bank_zero, CarrierReadout.bank_flow, CarrierReadout.bank_mem_heatDomain, CarrierReadout.heat_bank, CarrierReadout.bank_line, CarrierReadout.norm_bank_sq, CarrierReadout.chart_bank, CarrierReadout.LSeries_eq_chart_bank, CarrierReadout.laneChart_bank, CarrierReadout.lane_factor_ne_zero, CarrierReadout.zeta_zero_iff_bank_mem_ker`

Proof sketch. (i) is $x_n^{-(s+s')}=x_n^{-s'}x_n^{-s}$, and membership in the domain of $e^{-s'H}$ is (iii) at $s+s'$. (ii) is (i) at $s=\frac12$, $s'=it$, with $|x_n^{-it}|=1$. (iii): $|x_n^{-s}|^2=x_n^{-2\operatorname{Re} s}$, and $\sum_n n^{-p}$ converges if and only if $p>1$; the membership statement is `CarrierReadout.bank_mem`ℓ`p_iff` (the name carries a script ell). (iv) is a reindexing of the L-series. (v) is the pairing of consecutive phasors together with $\eta=(1-2^{1-s})\zeta$ on $\operatorname{Re} s>0$ (§6). (vi): in the open strip the lane factor $1-2^{1-s}$ does not vanish, since $|2^{1-s}|=2^{1-\operatorname{Re} s}>1$. □

(v) is (G1) made about the bank: the lane-balanced chart reads the object on the whole channel $\operatorname{Re} s>0$, where the plain chart reads nothing; and Theorem 17 lifts it to every carrier scale, to the three-dimensional phasors, and to the statement that the function has no content beyond the chart. (vi) is the dictionary between the chart and the geometry that §4 promised: a zero of the one-dimensional function is the H-flow of $\not\Vdash$ entering the kernel of the chart. And (iii) is the item we would point at. The bank is square-summable strictly to the right of the critical line and its norm blows up *at* the line: the line is where the Hilbert space in which the object lives ends. That is a property of the bank, established for every s. It says where the object lives, not where the chart vanishes.

Remark 64 (what this section does not say). No statement above locates a zero or constrains where one may be. Theorem 63(vi) is a dictionary, Theorem 63(iii) is a property of the bank, and the identities of Theorem 62 are Bost–Connes' partition function and the Euler product read on an operator. The spectrum of H is $\{\log x_n\}$, the sites; it is the *trace* of functions of H that carries the arithmetic, and the vector the chart projects that carries the geometry.

18 Where the geometry is load-bearing

The chart of the bank is Hardy's Z (Proposition 23), so everything the chart can say, the classical argument already says. This section is the inventory of what the chart cannot say: item by item, a property of the three-dimensional object, tied to a specific result that has no statement without it. Three of the nine are *negative* or costly — the geometry imposes a cost, or refutes something — and those are the ones we would defend hardest, because a frame that only ever agrees with the classical picture is decoration.

(G1) The lane split gives a series that converges *on* the line. The conductor sorts the fiber's phasors into $\pm$ channels; for ζ that is the non-principal character mod 2, and the lane-balanced sum is η, convergent for $\operatorname{Re} s > 0$. This is why the extractor can evaluate the object *on* the critical line by summation, with no analytic continuation and no functional equation in the evaluator. The isolation theorems do not need it — they work with Λ — but the *locator* does not exist without it. Used: every evaluation in §13.

(G2) The same lane split imposes a cost, and it is linear. The alternating sign that buys (G1) also manufactures a stationary point: the n-th term's phase is $\pi n - T\log(\Delta n)$, stationary at $n_* = T/\pi$, forcing a head $X > 2T/\pi$ — *linear* in the ordinate, against $O(\sqrt{T})$ for Riemann–Siegel. This is a geometric prediction with a measured consequence (§13), and it is a cost rather than an advantage. We report it as such, and (O4) is the workaround. The carrier itself supplies far more than the head consumes: by the ordinate y it has laid down $N(y) \geq (3r/p^2)e^{2y}$ sites (Proposition 5), so the head is a truncation of the geometry's supply, not a shortage of it.

(G3) The μ_6 cell width is what refutes the sector balance law. Theorem 13 is a theorem *about* the carrier: a sector is an arc of width $2\pi/6 = \pi/3$ — the μ_6 cell — and the obstruction is that such an arc cannot cancel against its own antipode, so differencing antipodal sectors adds the masses. The constant $\sqrt{3}/2 = \cos(\pi/6)$ in that bound is the cell half-width. Change the cell and the theorem changes; delete the geometry and there is no statement left to prove. Used: §8, and it is why the μ_6 machinery is not idle even though the balance law built on it is false. And the cell is the helix's, not a bookkeeping device: cell(n) is the angle of the helix at turn parameter $n/6$, six sites per turn, with the antipode half a turn up (Proposition 7).

(G4) The ordinate variable and the lane factor give the abscissa reading its dictionary *and* its calibration. Under $s = \frac{1}{2} + iz$ a zero $\beta + i\gamma$ becomes a node $\gamma - i(\beta - \frac{1}{2})$, so the imaginary part of an eigenvalue is the signed distance from the critical line (§14). That dictionary is the coordinate's, and the coordinate is Riemann's — we claim no credit for it (§19(v)). What *is* the carrier's is the calibration: the fiber carries the lane factor $1 - 2^{1-s}$, whose zeros sit on $\operatorname{Re} s = 1$ in closed form, supplying a known off-line node set at $\operatorname{Im} y = -\frac{1}{2}$ spaced $2\pi/\log 2$. An instrument claiming to measure the abscissa is then checkable against zeros whose abscissa is known and is not $\frac{1}{2}$, and it passes to 2×10^{-14}. Without the lane structure there is no such standard and the reading is unfalsifiable.

(G5) The double end is what produces the off-line pairing. The geometry has two ends, mirror images through the origin, and the readout cannot tell which it reads; in the ordinate variable that end-swap is $z \mapsto \bar{z}$ (Lemma 32). This is not an aside: it is the step that composes the two symmetries into one. The identity $\Xi(\bar{z}) = \overline{\Xi(z)}$ consumes *both* the conjugation symmetry of Λ_0 *and* its functional equation — neither alone suffices, precisely because $z \mapsto \bar{z}$ is the composite $s \mapsto 1 - \bar{s}$ — and from it follow the off-line pairing (Theorem 33) and, through it, the strip parity law (Theorem 34). Delete the double end and those two theorems have no proof here. The dependency is checkable in the sources, and one half of it is not in Mathlib at all: `xiEntire_conj` is proved from Mathlib's functional equation `completedRiemannZeta`$_0$`_one_sub` together with the conjugation symmetry `HelixCollapse.completedRiemannZeta`$_0$`_conj`, which is this paper's (§21: from the real Dirichlet coefficients on $\operatorname{Re} s > 1$ and the identity theorem), and `offAxis_pairs` is proved from `xiEntire_conj`. Nor are the two symmetries two in the

geometry: the end-swap is one involution — conjugation, phasor by phasor — and the functional equation is what the chart reads of it (§4, "the functional equation is the double end"). That is the claim: not that either symmetry is new, but that the pairing is the chart reading of one geometric involution, and that the proof here consumes exactly that composite.

(G6) The radial direction is what makes the band verification possible. This is the strongest item and it is §16. The descent of §5 discards the radius; what survives into the ordinate variable is a direction in which "off the critical line" reads as "off the real axis". A contour count over $|\operatorname{Im} z| \le h$ therefore counts the zeros with $|\beta - \frac{1}{2}| \le h$ — and matched against certified sign changes it certifies that no zero in a window has left the line by as much as h (Theorem 56), and, with the sliver near $\operatorname{Re} s = 1$ closed by an elementary certificate, that every zero with ordinate in the window lies on the line (Corollary 59). The sign-based chain provably cannot make that statement: the proof is Theorem 34, which says that a sign change registers the parity of the *full* strip count, so a conjugate off-line pair, which adds two, is invisible to it (§19(iv)). The band count sees it because the distance off the line survives the descent as a direction, $\operatorname{Im} z = \frac{1}{2} - \beta$ (equation (2)), and that direction is the object's: it is exactly what the chart of §5 dropped. Here the geometry is not restating an analytic argument in other words; it is supplying the coordinate in which the question is askable.

(G7) The rank carries a count, and the eigenvalues carry the abscissa. The band verification of (G6) counts by *contour*, which is Turing's quantity reached in a different coordinate. The carrier offers a second counter, and it is the more informative one: the bank's log-jets are a power sum over the fiber's zeros, so a Hankel matrix built from them has a spectrum whose *rank* counts distinct nodes and whose *eigenvalues* return those nodes — imaginary part included, which is the abscissa (§14). A contour count returns an integer; the pencil returns the integer *and* where the zeros are and how far off the line each one sits. Both halves are certified: the singular values by §14.7, and the rank itself by Theorem 50 once the moments are restricted to a contour — so this is a second certified counter, not merely a diagnostic.

(G8) The focal picture is a theorem, for characters the identification misses. Theorem 11 is an *iff* between a geometric event — the $\pm$ channels closing at a height — and the vanishing of an L-value, for non-principal real χ. It is genuine content about the model, and Proposition 23 does not dissolve it, because the identification with Z is specific to ζ's archimedean factor. What must be said, and is, is that it does not cover ζ: the $\pm$ split is by character *value* and degenerates at the trivial character (Remark 12).

(G9) The fiber has a Hamiltonian, and the critical line is where its Hilbert space ends. The warp's generator is an unbounded self-adjoint operator on $\ell^2(\mathbb{N}, \mathbb{C})$ with the phasors as eigenbasis (Theorem 60); the fiber is its flow, the bank $B(s) = e^{-sH}\mathbb{1}$ (Theorem 63); ζ is its trace, and the prime side of the explicit formula is the trace of a function of it (Theorem 62). Two consequences are the geometry's and not the chart's. The bank is square-summable exactly to the right of the critical line, $\|B(s)\|^2 = \zeta(2 \operatorname{Re} s)$, so the line is the boundary of the Hilbert space in which the object lives — a fact with no one-dimensional expression, because the one-dimensional function is a chart of the bank and not the bank. And every L-series is a chart of the *same* bank: the carrier-generic law of §1, in operator form. Used: §17; it is the reading under which the "generator" of §1 is a theorem. Its eigenvalues are the geometry's

twice over: the ordinates at which the climber reaches the sites (§4), and the only continuous weights for which the carrier scale is a gauge (Theorem 4) — the site weights are derived, not posited. Each of these is a compiled statement about the *bank*, not about the number $\zeta(s)$: `CarrierReadout.bank_mem`ℓ`p_iff` is the boundary, `CarrierReadout.LSeries_eq_chart_bank` the universality, `CarrierReadout.zeta_eq_heatTrace` the trace. A one-dimensional reader holding $\zeta(s)$ holds the value of one chart at one point; $\|B(s)\|^2$ is the value of *another* chart of the same vector, at $2\,\mathrm{Re}\,s$, and the statement that the vector leaves ℓ^2 at the line is about the vector.

What this does not amount to. Not that the geometry proves the isolation theorems — it does not, and Proposition 23 is the reason. What it does is supply the object that is computed (G1) and the cost of computing it (G2); the constant in a refutation (G3); a closed-form off-line calibration standard (G4); the involution behind two of the theorems (G5); the coordinate in which a statement the chart cannot make becomes provable (G6); a counter whose eigenvalues carry the abscissa (G7); one equivalence of its own (G8); and a Hamiltonian whose Hilbert space ends at the line (G9). Three of those nine are negative or costly, which is the test we would apply to any frame: one that only ever agrees with the classical picture is decoration.

19 What is not claimed

This section is not optional, and it is longer than the contribution on purpose.

(i) *No speed claim.* We do not claim to beat a state-of-the-art refinement, and an earlier draft of this work did, wrongly. Newton's method in ball arithmetic with precision doubling costs about two full-precision evaluations at any digit count, and it can share the same derivative vector this construction computes; measured against a correct baseline it wins. The jet enclosure's advantage is that it produces the enclosure and the ordinate from *one* pass over the bank, not that the resulting arithmetic is cheaper.

The 198× reported in §20(O4) is not a counter-example to this and should not be read as one. It compares the one-ended lane bank against a two-ended evaluator *within this implementation* — a statement about the two heads, not about this work against anyone else's. Against Platt's isolation or Newton refinement in ball arithmetic we make no comparison at all.

(ii) *Index to certified interval is not new.* Turing's method in Platt's rigorous form [12] does exactly this and is shipped in Arb [9]. Anyone who wants the i-th ordinate should call that. The claim here is a representation with a machine-checked path to $\zeta(\frac{1}{2} + i\gamma) = 0$, not a new capability.

(iii) *The jet enclosure is not a new localization principle.* It is the construction of Gargantini and Henrici [7] — locating a zero from the logarithmic derivative and finitely many of its derivatives at a point, in circular arithmetic — with the circular regions supplied by the carrier's tail bound. Their paper already applies the idea to transcendental functions.

(iv) *The index certifies less than it appears to.* What Theorem 45 returns — and it is the product theorem rather than the capstone that the implementation instantiates, see (O1) — is: an interval containing a zero of ζ on the critical line, given a count. It does not certify

that this is the i-th zero in the strip unless the count supplied is the strip count; with an on-line count it certifies the i-th *on-line* zero, and Theorem 34 is exactly the statement of how much the readout can see about the difference. The readout cannot distinguish an on-line zero from a conjugate off-line pair by its sign alone — that is the content of the parity law, not a defect of the implementation.

The qualification is Theorem 56: the *sign* cannot separate the two, but the band count can, and where both are certified the separation is established for that window — and the band itself is removable (Corollary 59). So the limitation is real for the chain of §§7–11 and is lifted, at a cost in evaluations, by §16.

(v) *The coordinate is Riemann's.* The substitution in which the functional-equation involution $s \mapsto 1 - s$ becomes complex conjugation is $s = \frac{1}{2} + iz$, which is Riemann's Ξ variable [2]. The carrier supplies the lane structure and the gauge; it does not supply that coordinate, and we do not claim it.

(vi) *The Riemann Hypothesis is not involved.* Nothing above assumes it, nothing above bears on it, and the count interface is agnostic to it: if a zero left the line the parity law of Theorem 34 is what would report it.

(vii) *The eigenvalue method of §14 is classical, and its rank does not see multiplicity.* Recovering nodes from a power sum is Prony's problem, and the matrix-pencil form used here is Hua and Sarkar's; neither is ours. What §14 supplies is the reading — that under $s = \frac{1}{2} + iy$ the fiber's log-jets are a power sum whose nodes carry the abscissa in their imaginary part, and that the lane factor furnishes a closed-form off-line calibration set. Two limits are stated there and repeated here because they are easy to overstate. The numerical rank counts *distinct* nodes, so a double zero is indistinguishable from a simple one and a rank drop reports coincidence rather than multiplicity. And the whole section is a measurement: the node errors are observed, not bounded a priori, so the abscissae it reports are evidence about particular zeros and not certificates about any of them.

(viii) *The focal equivalence is proved for non-principal real characters and does not extend to ζ in that form.* Theorem 11 — the $\pm$ channels closing at a height if and only if the L-value vanishes — carries the hypothesis $\chi \neq 1$, so it says nothing about ζ; the split there is by character *value*, and for ζ there is no non-trivial value to sort by. The sector statistic that replaces it for ζ is refuted in Theorem 13. What licenses the ζ chain is neither of these but the exact factorisation of Theorem 19, which is an identity.

(ix) *The Hamiltonian reading is Bost–Connes', and the bank's boundary is a property of the bank.* That ζ is the partition function of an operator with spectrum $\{\log n\}$ is [13]; §17 compiles it against this paper's operator and adds the bank, its unitary orbit on the line, its ℓ^2-boundary, and the reading of every L-series as a chart of it. None of that locates a zero. Theorem 63(vi) is a dictionary between the chart and the geometry, and Theorem 63(iii) says where the object lives; neither says where the chart vanishes, and we make no such claim.

With Theorem 13 in place, no statement in this paper is conditional on an unproved hypothesis, and no theorem cited here takes one as an argument. What remains is of a different kind and is itemised next, with a status line for each: the sign evaluations and the two-ended head are now

discharged by computation, the two error terms of (O3) are open but off the certification path by construction, and the count is open *by design*, since it is the interface of Definition 44 and certifying it is Turing's method. None of these is a gap in a proof.

20 Remaining obligations

The theorems above are complete as stated. This section is about the distance from those statements to a fully certified pipeline. It began as four open items; two of them are discharged here and the other two are reduced, so what follows records both what closed and what did not.

	status	where
(O1) which theorem the code instantiates	resolved: the product theorem	below
(O1) certified sign evaluations	**closed**	`certified_zeta.py`
(O1) certified *count*	**closed** via (O5)	`certified_count.py`
(O1) $\Xi' \neq 0$: the *capstone's* hypothesis	**closed** via (O5)	`certified_deriv.py`
(O2) a bound on S	**closed in the negative**	`Sx_bounded.py`
(O3) two error terms	resolved in role: evaluation count only	`certified_bisect.py`
(O4) two-ended head	**closed**, 198× measured	`two_ended_head.py`
(O5) explicit region bound on $\lvert\Xi\rvert$	**closed**: derived below	`region_bounds.py`

The structure of this section is (O5). Several obligations that looked independent all needed the same missing ingredient — an *explicit* bound on $|\Xi|$ over a complex region — and supplying it discharges them together. It is derived below from the same Euler–Maclaurin expansion the certified evaluator uses, so no constant is quoted that is not computed here. One item, (O2), does *not* reduce to it and remains open; it is a question about the distribution of zeros, and it bears on speed alone.

(O1) The implementation realises the product theorem, not the capstone, and its arithmetic is not certified. Two separate things, and an earlier draft of this section ran them together.

Which theorem the implementation instantiates. Theorem 42 needs $\Xi' \neq 0$ across each cell; Theorem 45 needs no derivative hypothesis at all, only the two endpoint counts and a sign bracket. The extractor supplies the counts (Stage 2) and the bracket (Stage 5), so what it instantiates is the product theorem. This is not a technicality, because the capstone's hypothesis is in general *false* on the bracket the extractor produces. Between consecutive zeros of Ξ there is necessarily a critical point, and a bracket a mean spacing wide generally reaches past one of the two flanking it. Measured directly (`extractor/checks/deriv_in_cell.py`): at $i = 10, 100, 1000$ the Stage-2 bracket contains exactly one sign change of Ξ' — at $i = 1000$, a critical point in $[1418.9845, 1418.9961]$ against $\gamma_{1000} = 1419.4225$ — while at $i = 1$ it contains none. On the narrow interval Stage 5 returns, Ξ' has no sign change at any index tested. So a capstone-shaped certification is available, but only with the cells taken to be the returned intervals rather than the search brackets, and the present code does not do that.

The arithmetic — now certified. The sign evaluations no longer rest on a noise floor. A certified evaluator is supplied in `extractor/certified_zeta.py`: the Euler–Maclaurin expansion

$$\zeta(s) = \sum_{n<N} n^{-s} + \frac{N^{-s}}{2} + \frac{N^{1-s}}{s-1} + \sum_{k=1}^{K} \frac{B_{2k}}{(2k)!}(s)_{2k-1} N^{-s-2k+1} + E,$$

carried with its classical explicit remainder $|E| \le \left|(s)_{2K+1}\frac{B_{2K+2}}{(2K+2)!}N^{-\sigma-2K-1}\right| \cdot \frac{|s+2K+1|}{\sigma+2K+1}$, plus an accumulated bound for the working-precision rounding. This returns $\zeta(s) \in v \pm \rho$.

A word on what that word carries. The *truncation* bound is a theorem. The *rounding* bound is the standard accumulation $M \cdot 10^{-D}\sum|\text{term}|$ over M correctly-rounded operations — an assumption about `mpmath`'s arithmetic, not a theorem about it, and it is the one place where "certified" below means *certified relative to a stated model of the library.* Substituting a ball-arithmetic backend would discharge that assumption and change nothing else. It is worth being exact about this because the same word is used in §21 for machine-checked Lean, where it means something stronger. Since $\operatorname{sign} u_\Delta = \operatorname{sign} Z$ with $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ and the two discarded factors strictly positive, the sign of the readout is proved wherever $|Z| > \rho$.

Run on the extractor's own output (`extractor/checks/certified_signs.py`), at $i = 1, 10, 10^2, 10^3, 10^4$, at all four points the product theorem consumes — the two bracket endpoints and the two ends of the returned interval — *every sign is proved*, with margins $|Z|/\rho$ between 7×10^{11} and 6×10^{36}, and the sign change across the returned interval is proved at every index. The enclosure was itself validated against an independent implementation: `mpmath`'s `zeta` falls inside the computed radius at every height tested.

What is *not* certified by this is the count, and we attempted it rather than deferring it. The natural route is the argument principle — Delves and Lyness [1] — since the winding number is an integer and so only has to be pinned to within half a turn. Walking the boundary of a rectangle and accepting a step whenever $\operatorname{Re}\big(\Xi(z_{k+1})\overline{\Xi(z_k)}\big) > 0$ is certified looks sufficient, because that condition places the increment in $(-\pi/2, \pi/2)$.

Remark 65 (a criterion that looks sound and is not). It is not sufficient, and the failure is quiet. The condition pins the increment in $(-\pi/2, \pi/2)$ *modulo* 2π; it does not exclude the path making extra full turns between the two samples. On the rectangle $[10, 23] \times [-\frac{1}{2}, \frac{1}{2}]$, which contains γ_1 and γ_2, one sample per side has *all four* increments pass the test and returns the count 0; the true value 2 appears only from four samples per side onward, as `extractor/checks/winding_unsound.py` records. A local test that passes at every step, with an answer that moves as the sampling is refined, is the signature: soundness cannot depend on the sampling density.

The repair is standard and needs one thing we do not have. If L is the step length and M' bounds $|\Xi'|$ on the step, then $|\Xi(z_k)| > LM'$ forces Ξ to stay in the half-plane through $\Xi(z_k)$, which excludes the extra turns. By Cauchy, $M' \le M/r$ with M a bound for $|\Xi|$ on a disc of radius r. So the criterion becomes sound as soon as M is available — and M is (O5).

That repair is carried out in (O5) below, and with it the contour route works: the count is certified without an external counter. Turing's method in Platt's form [12], shipped in Arb [9], remains the fast route and is what we would recommend in practice; the point here is only that the paper does not have to borrow one.

What a partial attempt showed. Certifying $\Xi' \neq 0$ on an interval by a Cauchy estimate — $|\Xi''| \le 2M(r)/r^2$ on a disc of radius r, hence no critical point where $|\Xi'(m)| > (2M(r)/r^2) \cdot$ halfwidth — reaches a ratio of only 0.27 to 0.38 on the Stage-2 bracket, at the best radius, uniformly across $i = 1, 10, 100, 1000$; and subdividing makes it *worse*, not better. That is the signature of the hypothesis failing rather than the bound being loose, and it is what led to the measurement above.

(O2) The count is $O(i)$ in the index, and a bound on S would make it $O(1)$. Stage 2 bisects on Count, and the cost of the count is what the head law sets: linear in the ordinate,

hence linear in the index up to logarithms. If one knew $|S_x - S| < 1$ — that the clock-model count and the true count never differ by a whole unit — the index could be read off the archimedean winding directly and the count would cost $O(1)$ in the index. Here S is as in Remark 40 and S_x is its Selberg approximation at cutoff x, the finite prime-clock sum

$$S_x(t) \;=\; -\frac{1}{\pi}\sum_{n\le x^3}\frac{\Lambda(n)\,w_x(n)}{\sqrt{n}\,\log n}\,\sin(t\log n),$$

with w_x the standard Selberg damping weights ($w_x = 1$ for $n \le x$, tapering to 0 by $n = x^3$). It is a trigonometric polynomial in t with frequencies $\log n$, so it is computable outright; the question is only how far it can stray from S. We measured that rather than assuming it: over 3000-point grids spanning the ordinates γ_{50}–γ_{70} and γ_{1000}–γ_{1015}, with the Selberg cutoff $x = 100$ (`extractor/checks/count_error_grid.py`), the maximum of $|S_x - S|$ was 0.502 and 0.518 respectively, with median 0.018 and 0.051. The excursions are not spread out: the grid points within 0.02 mean spacings of an ordinate carry median error 0.44 and 0.47 and supply every sample above $\frac{1}{2}$, while beyond 0.35 spacings the error never exceeded 0.075. So the difficulty is concentrated at the events, which is where S jumps.

This is a statement about *speed, not soundness*: Theorem 41 delivers the index with no bound on S whatsoever. But the hypothesis it asks for is not merely unproved. At a fixed cutoff it is false, and the reason is immediate.

Proposition 66 (the fixed-cutoff hypothesis fails)**.** *For each fixed x, $\sup_t |S_x(t) - S(t)| = \infty$. In particular $|S_x - S| < 1$ holds at no fixed cutoff.*

Proof. S_x is a finite trigonometric polynomial, so for every t

$$|S_x(t)| \;\le\; B(x) \;:=\; \frac{1}{\pi}\sum_{n\le x^3}\frac{\Lambda(n)\,w_x(n)}{\sqrt{n}\,\log n},$$

an explicit finite constant. Meanwhile S is unbounded. Unconditionally

$$S(t) \;=\; \Omega_{\pm}\Big((\log t/\log\log t)^{1/3}\Big),$$

by Selberg's method [11], in the sharpened form due to Tsang. Hence $|S_x(t) - S(t)| \ge |S(t)| - B(x)$ is unbounded in t. □

The constant is not large. Computed (`extractor/checks/Sx_bounded.py`): $B(10) = 2.17$, $B(100) = 12.43$, $B(1000) = 110.5$. So at the cutoff the implementation uses, any t with $|S(t)| > 13.43$ already violates the hypothesis; such t exist by the proposition, though they lie far beyond computational reach — the largest known values of $|S|$ are near 3.3.

What is within reach is the threshold that actually matters. Reading an integer count off $\vartheta(t)/\pi + 1 + S_x(t)$ requires $|S_x - S| < \frac{1}{2}$, not < 1, and that fails already:

window	height	median	max$\lvert S_x - S\rvert$
γ_{50}–γ_{70}	143	0.017	0.501
γ_{500}–γ_{520}	811	0.047	0.526
γ_{1000}–γ_{1015}	1419	0.051	0.518
γ_{3000}–γ_{3015}	3533	0.062	0.530
γ_{8000}–γ_{8015}	8148	0.057	0.562

Every window exceeds $\frac{1}{2}$, and the maximum drifts upward with height, as `checks/Sx_growth.py` records. So the clock model at a fixed cutoff is a locator, never a certified counter — which is how the paper has used it throughout, and why the count is stated as the interface of Definition 44.

What the correct statement would be, and why we do not need it. A bound of this shape requires the cutoff to grow with the height, $x = x(t)$; that is Selberg's formula, and its uniform error term is the difficult part. We do not prove one. Nor do we need one: the count is certified by contour in (O5)(a) below, and Turing's method in Platt's form [12] supplies a fast certified counter. (O2) asked for a shortcut past both, and Proposition 66 says the shortcut does not exist at fixed cutoff. That closes the item — in the negative, which is the only direction available.

(O3) Two error terms are measured and not proved. The finite-bank moment bound — the α, β of Theorem 43 as functions of the head length — is at present the empirical law of §13, $\exp(-cs)$ with $c \in [1.400, 1.485]$ measured across thirteen orders of magnitude of error at 60-digit working precision, uniformly in the moment order $r \leq 4$ and in the carrier scale. The proof route is stationary phase against the saddle at $n_* = T/\pi$, the same saddle the head law already identifies; what is missing is a remainder with explicit constants, not the shape. Separately, the derivative remainder in the Riemann–Siegel comparison is quoted rather than derived, and the route there is Gabcke's explicit error estimates [6], which supply exactly such constants for the first eleven partial sums at $t \geq 200$.

Both sit on the jet path, and with (O1) in hand their status is now structural rather than asserted. The certification runs through the Euler–Maclaurin evaluator, which carries its own explicit remainder; the lane bank's truncation error and the Riemann–Siegel remainder enter only the *locator* and the jet's ε. An unproved α, β therefore costs narrowness, never correctness — and that is now true by construction, because the object that decides the signs is not the object these two error terms belong to. What remains open in (O3) is consequently a question about performance parameters, not about soundness: the two constants govern how wide the returned interval is and how long the bracketing takes, and neither enters the proof that the interval contains the event.

(O4) The two-ended head, realised. The head law forces $X > 2T/\pi$, linear in the ordinate, and this is the construction's one structural cost. An earlier draft of this section said the two-ended form "has to be derived for the lane-balanced bank rather than borrowed". That was the wrong framing, and it is worth saying why, because the same observation is what closes the item: every compiled theorem consumes the readout *only through its sign*, and $u_\Delta = |w_\Delta|\,|\Gamma_{\mathbb{R}}|\,Z$ with both discarded factors strictly positive. So the sign may be taken from any evaluator of Hardy's Z — in particular from the Riemann–Siegel main sum, which is two-ended by construction and has $N = \lfloor \sqrt{t/2\pi} \rfloor$ terms. Nothing needs deriving for the lane bank; the lane bank is the locator, and the evaluator is interchangeable.

Realised in `extractor/checks/two_ended_head.py` as a vectorised `float64` main sum with the first Riemann–Siegel correction term, and measured:

T	one-ended X	two-ended N	terms ratio	2000 sign evaluations one-ended	two-ended
7.0×10^3	4891	33	148	0.219 s	0.064 s
5.76×10^5	367124	302	1216	13.06 s	0.066 s

A speedup of 3.4× at $T \approx 7 \times 10^3$ and 198× at $T = 5.76 \times 10^5$ — so this is now a wall-clock measurement and not merely a term count. Against the one-ended lane bank the two evaluators agreed on the sign at 400 probes at each of four heights, with zero disagreements; against `mpmath`'s `siegelz` the float64 value agrees to 3×10^{-5} at $T \geq 7 \times 10^3$.

Two honest qualifications. The term-count ratio is 1216 at $i = 3 \times 10^6$, not the 605 an earlier draft reported: that figure double-counted, taking two sums of length $\sqrt{T/2\pi}$ where the standard Riemann–Siegel main sum is one sum of that length with a factor 2. And the float64 main sum with one correction term carries error $\sim 3 \times 10^{-5}$, which decides the sign comfortably away from an event but not within the returned interval; the architecture that follows is therefore two-ended Riemann–Siegel for the bracketing, where nearly all evaluations happen, and the certified evaluator of (O1) for the final signs.

(O5) An explicit bound on $|\Xi|$ over a region. What is needed is a function M with $|\Xi(z)| \leq M$ on a disc, with explicit constants — not an asymptotic order and not a sampled maximum, both useless here because the conclusions drawn from M are certificates. It is supplied in `extractor/region_bounds.py`, and no constant is taken from the literature:

- $|\frac{1}{4} + z^2| \leq \frac{1}{4} + (|m| + r)^2$ and $|\pi^{-s/2}| \leq \pi^{-(1/2-r)/2}$, both elementary;
- $|\Gamma(s/2)|$ by Stirling with its explicit remainder. The familiar constant $1/(12|w|)$ is the *real-argument* form. For complex w the remainder is sector-dependent:

$$|R_1(w)| \;\leq\; \frac{\sec^2(\frac{1}{2}\arg w)}{12|w|}.$$

  On every box used here $\operatorname{Re} w > 0$ and $\operatorname{Im} w > 0$, so $0 < \arg w < \pi/2$ and $\sec^2(\frac{1}{2}\arg w) < 2$, giving the safe uniform bound $1/(6|w|)$ — which is what the code uses. The difference is numerically immaterial ($|w| \approx m/2$, so the term is $\sim 10^{-3}$ at $m = 14$), but the real-argument hypothesis is not the one that applies here and we do not invoke it. $\operatorname{Re}[(w - \frac{1}{2})\log w - w]$ is bounded over the box by its value at a corner, each product being linear in each factor;
- $|\zeta(s)|$ *term by term from the same Euler–Maclaurin expansion the certified evaluator uses.* This is the step that removes the need for a quoted subconvexity constant: every term, and the remainder, is an explicit function of (m, r, N, K), so bounding $\operatorname{Re} s$ below by $\frac{1}{2} - r$ bounds the whole sum.

The precondition $|h| + r < \frac{1}{2}$ on a disc centred at $m + ih$ is not a coding limit: it is exactly $\operatorname{Re}(s/2) > 0$, the hypothesis of the Stirling remainder. Validated as a genuine upper bound at seven (m, r) pairs from $m = 14$ to $m = 1419$ (`extractor/checks/region_bounds_val.py`); it exceeds the sampled maximum of $|\Xi|$ every time, by factors between 25 and 520. Looseness of that order costs step size, not validity.

Three things follow, by Cauchy's estimate $|\Xi^{(k)}| \leq k!\, M/r^k$.

(a) The count, certified. The criterion of Remark 65 becomes sound in the form: a step of length L is safe when $|\Xi(z_k)| > L \cdot M/(R - L/2)$, which forces Ξ into an open half-plane along the step and so admits no hidden turn. Run on the band $|\operatorname{Im} z| \leq \frac{1}{4}$ with $R = 0.18$ (`extractor/checks/certified_count.py`), it returns the correct count on all six test rectangles

— including [10, 23], the one the unsound criterion got wrong. The cost is real: 10^4 to 4×10^4 evaluations per rectangle, because steps must shrink where $|\Xi|$ does.

(b) The capstone's hypothesis, certified. On $[\ell, h]$ the mean value theorem gives $|\Xi'(t)| \geq |\text{slope}| - \max|\Xi''|(h - \ell)$ with $\max|\Xi''| \leq 2M/r^2$, so $\Xi' \neq 0$ there whenever the slope's certified lower bound exceeds $2M(h - \ell)/r^2$. On the intervals the extractor returns this holds at $i = 1, 10, 10^2, 10^3, 10^4$ with margins from 10^4 to 2×10^8 (`extractor/checks/certified_deriv.py`). So the capstone's hypothesis — not merely the product theorem's — is discharged exactly where (O1) showed it is true.

(c) (O3), resolved in role. With signs certified, bisection alone narrows the enclosure. The script `extractor/checks/certified_bisect.py` drives it from certified signs and reaches widths of 10^{-20} down to 10^{-35} in 34 to 76 steps, stopping at the *evaluator's* precision, never at the bank's. The jets are therefore an optimisation on the number of evaluations, and their α, β govern neither soundness nor the achievable narrowness. Certifying them by contour, as Delves–Lyness [1] intended, is now unblocked by M and would buy speed; we have not implemented it, and nothing rests on it.

Relation to (O2). (O2) does not reduce to (O5) — it is a question about the distribution of zeros, not an estimate on $|\Xi|$ — and it is resolved separately, in the negative, by Proposition 66. The two together leave the certified count resting on (O5)(a) or on an external counter, and on nothing else.

21 Formalization

Every statement above carries a LEAN: line naming the identifiers that check it; those names are the statements' addresses in the sources and can be looked up verbatim. The modules divide along the same seam as the exposition: the geometry and its descent on one side, the chart chain on the other.

layer	module	theorems
geometry (§§2–3)	`ClosedForm.lean`	102
fiber and descent (§§4–5)	`GeometricReadout.lean`	18
	`CarrierParityQuantization.lean`	16
	`CarrierSignFlip.lean`	13
	`CarrierOffLinePairing.lean`	12
the chart chain (§§6–11)	`CarrierIntervalCount.lean`	14
	`CarrierStripParity.lean`	21
	`ZeroEnclosure.lean`	8
	`CarrierUniqueEvent.lean`	16
	chain total	100

The counts are the size of the artifact and nothing more: they are given so that a reader can find the sources and check that a cited name exists, not as a measure of content, and several of the counted theorems are one-line facts. The geometry modules are what make §§2–5 statements rather than a picture: the helix and its velocity, the closed-form arclength, the climber and the height identity $z = e^y$, the carrier index $N(y) = S(k(y))/\Delta$, the Cayley step, the global

inverse of the height readout, the antipodal cancellation of the μ_6 cell, the residue law, and Theorem 11 — the bridge identifying focal $\pm$ closure with the vanishing of the L-function — are all machine-checked, with the same three-axiom footprint as the rest.
Ten further modules are cited above but are not part of that chain and are not counted in the table:

module	theorems	carries
`HelixCollapseReality.lean`	6	reality of $\Lambda(\frac{1}{2}+it)$
`HingeKernel.lean`	14	Proposition 54
`UnitMidpoint.lean`	8	
`ZetaSectorBank.lean`	11	the μ_6 sector decomposition, Remark 12
`SectorBalanceRefuted.lean`	11	Theorem 13
`LogWeightOperator.lean`	20	Theorem 60
`CarrierHamiltonian.lean`	18	Theorems 61, 62
`CarrierBank.lean`	14	Theorem 63
`CarrierRegistration.lean`	11	Theorem 4, Propositions 7, 5, §4
`FiberIsTheFunction.lean`	10	Theorem 17

That last module is worth a sentence. `ZetaSectorBank.AngleSectorBalanceLaw` is a `def ... : Prop` — a named hypothesis, not a theorem — and one result in that module takes it as an explicit argument. The refutation module proves it false, unconditionally: `not_sectorBalanceAt` holds at every real t with no hypothesis at all, so that conditional result is vacuous rather than useful. After it, **no theorem cited in this paper takes an unproved proposition as an argument.**
Zero `sorry`, zero `axiom`. Every one of the 167 identifiers named in a LEAN: line above was checked *individually* with `##print axioms`, and each reports exactly

$$\{\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}\},$$

the three axioms of Lean's standard classical foundation, and nothing else. The check was run alongside a positive control (a deliberately incomplete proof in the same file), which reported `sorryAx`; without that control the audit would not establish that the procedure can fail.

What is taken from outside. The construction rests on the following facts from Mathlib. An earlier draft listed four and said "and on no others"; that was wrong — the boundary proposition (Proposition 35) consumes the non-vanishing of ζ on $\operatorname{Re} s = 1$, which is not a formality, and several smaller facts are used besides. The list below was extracted from the sources rather than recalled:

identifier	content
`completedRiemannZeta`	the completed zeta Λ
`completedRiemannZeta`$_0$	its entire companion Λ_0
`completedRiemannZeta_one_sub`	the functional equation $\Lambda(1-s)=\Lambda(s)$
`differentiable_completedZeta`$_0$	Λ_0 is entire
`completedRiemannZeta_eq`	$\Lambda=\Lambda_0-\frac{1}{s}-\frac{1}{1-s}$
`riemannZeta_def_of_ne_zero`	ζ from Λ off the poles
`Gamma`$\mathbb{R}$	the archimedean factor
`Gamma`$\mathbb{R}$`_ne_zero_of_re_pos`	$\Gamma_{\mathbb{R}}\neq 0$ for $\operatorname{Re} s>0$
`riemannZeta_ne_zero_of_one_lt_re`	$\zeta\neq 0$ for $\operatorname{Re} s>1$
`riemannZeta_ne_zero_of_one_le_re`	$\zeta\neq 0$ for $\operatorname{Re} s\geq 1$
`zeta_eq_tsum_one_div_nat_add_one_cpow`	$\zeta(s)=\sum_n (n+1)^{-s}$, $\operatorname{Re} s>1$
`LSeries_vonMangoldt_eq_deriv_riemannZeta_div`	$\sum_n \Lambda(n)n^{-s}=-\zeta'/\zeta$, $\operatorname{Re} s>1$
`riemannZeta_eulerProduct_tprod`	the Euler product, $\operatorname{Re} s>1$
`Real.log_nat_eq_sum_factorization`	$\log n=\sum_p v_p(n)\log p$
`LinearPMap.adjoint_isFormalAdjoint`	the adjoint of an unbounded operator
`HilbertBasis.mk`	orthonormal and total $\Rightarrow$ Hilbert basis
`Real.summable_nat_rpow_inv`	$\sum_n n^{-p}<\infty \iff p>1$

The substantive one among the first group is the non-vanishing on the line $\operatorname{Re} s=1$ (the last row of that group), a theorem of Hadamard and de la Vallée Poussin on which Proposition 35 rests. The second group serves §17 only, and is why Theorem 62 is described there as Mathlib's analytic facts read on an operator.

In particular the reality of $\Lambda(\frac{1}{2}+it)$ — Proposition 29, on which the whole readout depends — is *not* assumed. It is proved here, in a further module `HelixCollapseReality.lean`, from the functional equation together with the Schwarz-reflection symmetry $\overline{\Lambda(\bar{s})}=\Lambda(s)$: on the line $\overline{\frac{1}{2}+it}=1-(\frac{1}{2}+it)$, so the two symmetries compose to $\overline{\Lambda}=\Lambda$. The reflection lemma is not in Mathlib and is proved from the real Dirichlet coefficients on $\operatorname{Re} s>1$ and extended by the identity theorem. It carries the same three-axiom footprint.

Versions. The statements above are reproducible against the following revision, and are not claimed against any other.

Lean 4 toolchain	`leanprover/lean4:v4.34.0`
Mathlib tag (`inputRev` in `lake-manifest.json`)	`v4.34.0`
Mathlib commit (the tag's target)	`5ed2965256430c3649e86755f9576b54eca72435`

22 Reproducibility

The proofs. With the toolchain and revision of §21 in place,

```
lake build RequestProject.CarrierUniqueEvent RequestProject.CarrierStripParity
                                RequestProject.CarrierBank
```

compiles the two capstone modules and, transitively, the other five; the third target compiles the operator modules of §17. The axiom audit is a separate file of `##print axioms` lines carrying the positive control, run with `lake env lean`; it is not part of the library build, so that the control's `sorry` never enters the tree.

The extractor. `helix_extractor.py` takes indices on the command line and returns intervals:

```
python3 helix_extractor.py 1 10 100 1000 --verify
```

with `--delta` setting the carrier scale, `--tol` the head tolerance, `--x` an explicit head length, `--m` the jet order, and `--scales` running the carrier-scale invariance check of §13.3. The validation harness `extractor_validate.py` reproduces the table of §13.3 in full: 297 indices, 0 failures. Its reference ordinates come from `mpmath`'s `zetazero`, which shares no code with anything in this paper.

The numerical checks. Each measurement quoted above is a separate script with its decision rule fixed before it ran, so that a reader can see which outcome would have refuted the claim rather than only the outcome that occurred.

script	claim tested	pre-registered refutation
`checks/theta_identity.py`	$\arg\Gamma_{\mathbb{R}}(\frac{1}{2}+it)=\vartheta(t)$	any sample $> 10^{-30}$
`checks/jet_sweep.py`	contraction $=\kappa$	disagreement $> 5\%$ at small κ
`checks/count_error_grid.py`	size of $\lvert S_x - S\rvert$	(measurement; no threshold)
`checks/root_number_rays.py`	the χ_5, χ_7 rays	either ray off by $> 5\times10^{-4}\pi$
`checks/balance_law_refuted.py`	imbalance grows, ratio $> \sqrt{3}/2$	any decay toward 0
`checks/head_law_decay.py`	$-\ln(\text{err})/s$ band	(measurement; band reported as found)
`checks/head_law_moments.py`	band uniform in $r \le 4$	spread > 0.25 across orders
`checks/head_threshold.py`	$X_{\min}$ vs $2T/\pi$	(measurement; no threshold)
`checks/deriv_in_cell.py`	$\Xi' \neq 0$ on the returned interval	any sign change of Ξ' in it
`checks/certified_signs.py`	every sign proved, margin > 1	any point with $\lvert Z\rvert \le \rho$
`checks/two_ended_head.py`	two-ended signs agree	any sign disagreement in 400 probes
`checks/winding_unsound.py`	naive winding test unsound	a count stable under refinement
`checks/band_verification.py`	band count = real-zero count	a window where they differ
`checks/region_bounds_val.py`	M is an upper bound	any sampled $\lvert\Xi\rvert$ above M
`checks/certified_count.py`	counts match, soundly	any rectangle miscounted
`checks/certified_deriv.py`	$\Xi' \neq 0$ certified, margin > 1	any index with margin ≤ 1
`checks/certified_bisect.py`	narrowing is evaluator-limited	narrowing stalling at the bank's floor
`checks/rank_certified.py`	contour Hankel has rank K	rank above K, or s_0 non-integral
`checks/pencil_verify.py`	proposals verify at the anchors	any proposal unverifiable
`checks/certified_rank.py`	singular values certified (Weyl)	a rank-gap enclosure containing 0
`checks/strip_sliver.py`	the sliver is zero-free	a node with $\lvert\zeta\rvert - \text{rad} \le L\rho$
`checks/operator_positivity.py`	inertia reads 0, real windows	a certified negative eigenvalue
`checks/positivity_control.py`	planted pair: $\lambda_{\min} < 0$	a planted pair with $\lambda_{\min} \ge 0$
`checks/Sx_bounded.py`	S_x bounded by $B(x)$	any $\lvert S_x\rvert$ exceeding $B(x)$
`checks/Sx_growth.py`	$\max\lvert S_x - S\rvert > \frac{1}{2}$	a window staying below $\frac{1}{2}$
`checks/pencil_lane_control.py`	lane nodes at $\operatorname{Im} y = -\frac{1}{2}$	misplaced, or $\operatorname{Im} y = +\frac{1}{2}$
`checks/rank_drop.py`	rank = #distinct nodes	σ_2/σ_1 stays $O(1)$ through a merger
`checks/lehmer_rank.py`	the pair resolves at rank 2	one node returned, or $\beta \neq \frac{1}{2}$
`checks/sx_formula.py`	printed S_x = the code's	any height differing by $> 10^{-12}$
`extractor_validate.py`	interval contains γ_i	any index missing its ordinate

Two of these did return a refutation, and both are reported above rather than discarded: the jet sweep at $\kappa = 0.882$ (§12), where the geometric law fails because no nearest ordinate dominates; and the planted-node control on the harmonic pencil (§14), which refuted two successive implementations whose own certificates looked clean. The S measurement is reported as a measurement in two windows at one cutoff, with no uniformity claimed.

23 What the exercise settled

The construction is a geometric model that computes, and the two halves of that are worth separating in closing.

As a model, its content is that the one-dimensional L-function is a chart readout of a three-dimensional object, and that the classical difficulties sort cleanly on that distinction. The abscissa of convergence and the strip are features of the projection; the double end, the focal cancellation and the height are features of the object. The sorting is not a matter of taste — the Cayley step is what makes the strip the image of a circle, and it is machine-checked, as are the helix, its arclength and the height identity $z = e^{y}$.

As geometry doing work, the accounting is §18: the chart of the bank is Hardy's Z up to a positive factor, so the isolation chain is what the chart reads and coincides with the classical argument, and the inventory there is of what the chart cannot say. Six of the nine items are ordinary — an object, a cost, a constant, a calibration, an involution, an equivalence. Three are not: in the ordinate variable the distance off the line is a radial coordinate, and a certified band count matched against certified sign changes yields a statement about β that the readout's sign provably cannot reach (Theorem 56, band-free by Corollary 59); the carrier's moment data carries a second counter, whose eigenvalues return the abscissa directly, whose singular values are certified (§14.7) and whose inertia reads the off-line count (§14.10); and the fiber's generator is a self-adjoint operator on ℓ^2 whose flow is the bank and whose Hilbert space ends at the critical line (§17). Those are the places where the three-dimensional frame is not a way of speaking about a one-dimensional argument.

As a computation, the chain from the representation to $\zeta(\frac{1}{2}+i\gamma) = 0$ is machine-checked end to end, the implementation is validated at 297 indices against an independent reference, and its own hypotheses are discharged against an explicit remainder rather than a noise floor. What it does not do is compete: Turing's method in Platt's form is faster and we say so repeatedly.

The part we would point a reader at, though, is what *failed*. Three propositions that this work carried at various times as plausible or open turned out to be false, and all three fell to the same observation — a finite object cannot track an unbounded one. The μ_6 sector balance law fails because a $\pi/3$ arc cannot cancel against its own antipode, so differencing antipodal sectors adds the masses (Theorem 13). The natural winding criterion fails because $\operatorname{Re}\big(\Xi(z_{k+1})\overline{\Xi(z_k)}\big) > 0$ pins the argument only modulo 2π, and the give-away was an answer that moved under refinement while every local test passed (Remark 65). The clock bound $|S_x - S| < 1$ fails at any fixed cutoff because S_x is a finite trigonometric polynomial and S is not bounded (Proposition 66). Each was more useful refuted than it would have been left open, and each cost less to refute than to keep.

References

[1] L. M. Delves and J. N. Lyness, *A numerical method for locating the zeros of an analytic function*, Math. Comp. **21** (1967), no. 100, 543–560.

[2] B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Königl. Preuss. Akad. Wiss. Berlin (1859), 671–680.

[3] A. M. Turing, *Some calculations of the Riemann zeta-function*, Proc. London Math. Soc. (3) **3** (1953), 99–117; doi:10.1112/plms/s3-3.1.99.

[4] D. H. Lehmer, *On the roots of the Riemann zeta-function*, Acta Math. **95** (1956), 291–298; doi:10.1007/BF02401102.

[5] Y. Hua and T. K. Sarkar, *Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise*, IEEE Trans. Acoust. Speech Signal Process. **38** (1990), no. 5, 814–824; doi:10.1109/29.56027.

[6] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann–Siegel-Formel*, Dissertation, Georg-August-Universität Göttingen, 1979. doi:10.53846/goediss-5113.

[7] I. Gargantini and P. Henrici, *Circular arithmetic and the determination of polynomial zeros*, Numer. Math. **18** (1971), 305–320.

[8] F. Johansson, *Rigorous high-precision computation of the Hurwitz zeta function and its derivatives*, Numer. Algorithms **69** (2015), no. 2, 253–270; arXiv:1309.2877.

[9] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292; doi:10.1109/TC.2017.2690633.

[10] A. M. Odlyzko and A. Schönhage, *Fast algorithms for multiple evaluations of the Riemann zeta function*, Trans. Amer. Math. Soc. **309** (1988), no. 2, 797–809.

[11] A. Selberg, *Contributions to the theory of the Riemann zeta-function*, Arch. Math. Naturvid. **48** (1946), no. 5, 89–155; reprinted in *Collected Papers*, Vol. I, Springer, 1989.

[12] D. J. Platt, *Isolating some non-trivial zeros of zeta*, Math. Comp. **86** (2017), no. 307, 2449–2467.

[13] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. (N.S.) **1** (1995), no. 3, 411–457.

[14] M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Rev. **41** (1999), no. 2, 236–266.

[15] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), no. 1, 29–106.